



Technology Support for the Rural Transportation Incubator

Final Report

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- John Taylor, Chief of Specialized Transportation Division, Pennsylvania Department of Transportation
- Sam Purington, Executive Director, Volunteer Transportation Center, Inc.

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Introduction

In the last decade, information technology has taken center stage in the world of personal transportation. The rise of mobile device-based platforms to enable a new sharing economy for mobility has disrupted both the transit and personal car ownership sectors. While large urban areas have been the primary targets of this disruption, no corner of personal mobility has been left untouched. Rural areas are no exception.

In the same period, the impact of poor mobility options for people living in rural areas has gained greater attention. One cause is the increase in the number of older adults as a percentage of the population as whole — an increase that is happening at a faster pace in rural areas.

With the convergence of these two highly impactful phenomena, one technological and the other demographic, it's no surprise that many are looking to technology for solutions to the challenges of providing mobility options in areas where needs are increasing. However, the variety, scale, and complexity of the challenges for rural mobility, combined with the immaturity, complexity, and urban focus of many of the proposed technologies means that solutions may not be as close at hand as first appears.

The question of how to best apply resources, technological or otherwise, to mobility in the rural context is not new. Partially in recognition of the need for transit options that respond to barriers encountered by older adults and people with disabilities, transit and social service providers developed the concept of mobility management, “a transportation strategy that focuses more on the customer and their needs, and the meeting of these needs through the coordinated use of a variety of providers.”¹ This approach has taken hold in rural areas, where many agencies now fund mobility manager positions to develop and support customer-focused services.

Purpose of Project

This report seeks to provide guidance to Arizona stakeholders for moving productively through a largely uncharted domain of applying technology to mobility in rural communities. It offers a set of pathways by which technology can be successfully deployed by transportation providers to address the unique needs for mobility by rural communities with a set of concrete action steps to build capacity and empower transportation providers with respect to technology, while using a mobility management lens that takes into account the value of coordination of services.

The project is part of the Rural Transportation Incubator, launched with support from The Harry and Jeanette Weinberg Foundation and other partners through Age Friendly Arizona. Through the incubator, four communities are launching new or expanded transportation services, technology solutions will be explored to make these services more effective, and a tool kit will be shared to encourage other communities in the country to replicate and adapt this work.

¹ KFH Group, “Expanding Access to Our Communities: A Guide to Successful Mobility Management Practices in Small Urban and Rural Areas.” Project 20-65 Task 68. Washington, DC: National Cooperative Highway Research Program, 2018.

Methods

This report draws on a range of sources to arrive at its conclusions:

- A review of relevant literature and academic research to develop historical context and glean prior lessons learned
- Interviews with industry leaders around the country to understand the current landscape of the public transportation sector and how other states and regions have brought technology to bear in support of rural mobility
- In-person and phone interviews with a range of Arizona leaders working in rural transportation to collect a sample of goals and aspirations of agencies, gather perspectives about the current conditions in the state of Arizona as a whole, and get a snapshot of agencies' capacities and daily operations
- A mobility management workshop conducted with rural transportation leaders from around the state
- The expertise of Kevin Chambers, Principal and Founder of Full Path, and author of this report.

The Problem

High and Growing Demand for Transportation

Rural America is experiencing a demographic shift that is already significantly altering how people are able to get around, and this shift will continue for decades to come. Nationally, people 65 years or older account for 17.5% of the rural population, compared to 13.8% in urban areas.² By 2030, people 65 years or older are projected to account for 20% of the total US Population, resulting in nearly a quarter of rural residents being of retirement age.³ While 90% of people over 65 prefer to stay in their home for as long as possible,⁴ the barriers to aging in place in rural areas are higher than in cities, due to more difficult access to services and a greater likelihood of reliance on a personally owned vehicle, reliance that can easily be undermined by disability.

Challenges Applying Technology

There are high expectations that technology will be able to mitigate the impact of this shift. However, several barriers are important to recognize and plan for in the effort to have technology realize its full potential for rural mobility.

² Smith, Amy Symens, and Edward Trevelyan, "The Older Population in Rural America: 2012–2016," ACS-41, American Community Survey Reports, U.S. Census Bureau, Washington, DC, 2019.

³ Colby, Sandra L. and Jennifer M. Ortman, "Projections of the Size and Composition of the U.S. Population: 2014 to 2060, Current Population Reports", P25-1143, U.S. Census Bureau, Washington, DC, 2014.

⁴ Farber, Nicholas, and Douglas Shinkle, "Aging in Place: A State Survey of Livability Policies and Practices," AARP Public Policy Institute, 2011.

Perhaps the most visible generator of these high expectations for technology is transportation network companies (TNCs) such as Uber and Lyft, which have gotten attention as possible rural mobility solutions. While some transit agencies have piloted public-private partnerships with them, and the influence of these businesses is likely to increase over time, it remains unclear in what scenarios TNCs will be able to reach and sustain profitability. Beyond questions about how well TNCs will be to support transit overall, there is an extra level of doubt as to whether a service that developed in urban spaces is the proper match to rural settings. Rural areas present a particular challenge compared to cities due to reasons that include poor internet access, lower penetration of credit cards, and low driver payout.⁵ One case study of small urban agencies using a TNC as a form of transit found problems with driver availability.⁶

Apart from these recent entrants, the vast majority of the lower-profile but decades-long efforts to bring effective technology into rural transit has been made by public transportation providers and human services agencies who have built websites; purchased off-the-shelf software for scheduling, dispatch, and asset management; and installed hardware in vehicles for driver communications and relaying vehicle location in real time. While this approach has been successful in a great many instances, success has largely depended on factors that put smaller and more rural operations at a disadvantage. In one of the few profiles found in the literature that recounted an innovative rural transit technology implementation from the perspective of a rural transit agency, the implementing agency found “there was not time or resource for a small frontier rural agency to undertake [a] time-consuming complex use case document”⁷ as is commonly done to successfully carry out such projects.⁸ This lack of resources appears to have been a contributing factor to the project not being fully implemented.

One of the areas where technology is often expected to add value is with respect to the coordination of different mobility options. This desire has spurred the federal government to fund pilots and research⁹ into the creation of “travel management coordination centers” (TMCCs) charged with being the hub for coordinating specialized transit in a region. The Reference Manual for Planning and Design of a TMCC¹⁰ recognizes that “the coordination of services for the transportation-disadvantaged is a very complex issue with many dimensions of policy, institutional, organizational, and human challenges” while also seeing technology as a potential enabler of coordination. In order for technology to fulfill its potential, the manual calls out a set of

⁵ Shrikant, Aditi. “Transportation Experts See Uber and Lyft as the Future. But Rural Communities Still Don’t Use Them.” Vox. VoxMedia, January 11, 2019. www.vox.com/the-goods/2019/1/11/18179036/uber-lyft-rural-areas-subscription-model.

⁶ Todd Hansen, Zachary Elgart, Kristi Miller, and Maarit Moran, “Existing Transportation Network Companies Used as a Part of Basic Mobility: White Paper”, Texas A&M Transportation Institute, November 2018.

⁷ A use case document is a thorough description of how a technology system and its users interact to accomplish a particular outcome. For example, use case document could describe the case of how a scheduler uses with a scheduling and dispatch application for assigning the next day’s ride requests to drivers and vehicles.

⁸ Couch, Pamela and Marcy Jaffe, “Evolution of Intelligent Transportation Systems for Mobility Management and Coordination Serving California’s Rural Frontier”, FTA Research Report No. 0006, Federal Transit Administration, 2012.

⁹ A primary vehicle for this research has been the Mobility Services for All Americans (MSAA) grant program. Details can be found at the Intelligent Transportation Systems Joint Program Office of the U.S. Department of Transportation: www.its.dot.gov/research_archives/msaa/index.htm.

¹⁰ Hemily, Brendon, “Reference Manual for Planning and Design of a Travel Management Coordination Center (TMCC)”, FTA Research Report No. 0117, Federal Transit Administration, 2017.

design and planning steps called *systems engineering* as not only a practical tool that is key to success in a complex project, but also a requirement for many federal grant programs. A systems engineering approach, while not absolutely beyond the reach of rural providers, does require significant time and resources to apply well. This helps to explain the relative scarcity of robust coordination systems being successfully developed in rural areas, where resources are already tight.

In addition to the mismatch between agency capacity and the level of effort that best practices like systems engineering recommend when implementing technology, there is a similar mismatch in terms of agency expectations. Agency decision makers commonly expect technology made for transit and community transportation to work much more smoothly and intuitively than it usually does. Overly optimistic vendor sales pitches are one source of this misconception. Another is unconscious assumptions rooted in the experience of using consumer-grade technology such as Microsoft's Office suite, mobile devices, or Google Maps. However, similar to specialized technology in other fields, systems designed specifically for smaller transit systems or volunteer driver programs differ from consumer technology in two fundamental ways:

- They are usually less polished, less stable, and more difficult to use. This is due to having a much smaller user base and fewer resources available to improve the product over time. The practical impacts are lower reliability, higher risk of problems over the useful life of the system, and diminished user friendliness. On the one hand, the systems can be subject to more disruptive change as developers work towards improving them while lacking the resources needed to make the updates smooth for agencies. On the other, they can fail to keep pace with larger technology trends due to the vendor working in too small a niche to have the means to upgrade the inner workings of their solution.
- They are often closely tied to organizational workflows. Specialized technology often comes with impactful yet unspoken assumptions and opinions about its place in an organization's inner workings. As one director of a volunteer driver program put it, "they force you into their methodology."¹¹ Software is often procured under the assumption



Figure 1 — While generalized technology like fax machines and email tend to be low-risk investments, specialized technology that is tied to complex workflows can be more difficult to implement and maintain than is initially apparent. With so many parts and people involved, they can resemble Rube Goldberg machines, like this apparatus to sharpen a pencil

¹¹ Interview with Sam Purington, Executive Director of Volunteer Transportation Center, Inc, November 19, 2019.

that all agencies work or should work basically the same way, or that the software vendor has found the most correct way to do things.

Much of the research that touches on the application of technology by transit agencies does not include the voices of agency staff. Where those voices are absent, there appears to be a tendency to understate the level of effort required to deploy systems. In reports that include viewpoints of agency staff, the picture becomes clearer. One report for a coordination project, which does include a perspective from agency staff, recommends that others “double the training budget” and “secure a systems engineer, if possible”.¹² Another points out that “technology for the urban context cannot be readily transferred to the rural context with the assumption that there will simply be fewer trips.”¹³

Taken together, these mismatches lead to industry-wide problems: overestimating the value that technology can bring to rural mobility, underestimating the organizational capacity required to implement specialized technology, and taking on greater risk in projects than is planned for.

Finally, an inflated understanding of technology’s capabilities can eclipse other worthy investments that can also improve mobility, access, and the social determinants of health for rural communities. For example, the rules of geometry and travel times mean that even a perfect vehicle schedule optimization algorithm can only marginally improve the productivity and cost effectiveness of a door-to-door service in a rural or low-density setting. Practical constraints such as this are common knowledge within the transit field, but in recent years they have become blurred in the public discourse.

Amid the recent explosion of mobility startup enterprises such as TNCs, some have claimed that app-based flexible services can be a sort of silver bullet by overcoming the trade-offs that are baked into the challenge of moving people.¹⁴ Given this current over-promotion of technology, it is worth noting that while applying it can provide critical support to help an expanding service manage its growing pains, it is more limited in what it can do to make a small, static service become more efficient. For example, a volunteer driver program with a dozen volunteers implementing a full-featured scheduling and dispatch system is unlikely to experience benefits outweighing the costs of putting such a system in place. For this reason, a potential application of technology should be carefully analyzed for return on investment alongside others possible uses of resources, such as purchasing additional vehicles and adding routes, as well as non-transit efforts that have effect of improving mobility or other social determinants of health, such as building sidewalks or investing in rural medical facilities.

¹² Basshgam, Lynnda and Dana Luttrull, “Lower Savannah Aging, Disability & Transportation Resource Center: Regional Travel Management and Coordination Center (TMCC) Model and Demonstration Project”, FTA Research Report No. 0065, October 2014.

¹³ Couch, Pamela and Marcy Jaffe, “Evolution of Intelligent Transportation Systems for Mobility Management and Coordination Serving California’s Rural Frontier”, FTA Research Report No. 0006, Federal Transit Administration, 2012.

¹⁴ Walker, Jarrett. “Microtransit: What I Think We Know.” Human Transit. Jarrett Walker Associates, June 24, 2019. humantransit.org/2018/02/microtransit-what-i-think-we-know.html.

Findings

Arizona

These findings are drawn from an evaluation of Arizona's rural specialized transportation system. The system evaluation is presented in the format of a SWOT analysis and can be found in Appendix B of this report.

Arizona is a state with large rural areas, a high level of demand for increased mobility by residents living in those areas, and a limited amount of state-level funding to address that need. In addition to those challenges, which exist in other states,¹⁵ interviewees in Arizona cited that they see a significant number of older adults moving to the state to retire without the benefit of extended family or a well-developed support network. Interviewees believed that this in-migration may be resulting in demand for public and community-based transportation services that are higher than that found in other states.¹⁶

System Evaluation – SWOT Analysis

Strengths	• Regional mobility manager network • Services rooted in community needs • Agencies success with technology • Interest in coordination • MAG capacities	Weaknesses	• Significant isolation for some providers. • Insufficient state-level coordination to apply technology
	Opportunities • RTI project • Possible alliances • Software-as-a-service • Tech tools for service discovery		Threats • Demographic trends and land use patterns • Cultural opposition to transit • Flat or declining funding environment • Limits of technology

Figure 2 - Summary of SWOT Analysis. Full details are in Appendix B.

Despite these challenges, visits and interviews with a selection of Arizona transportation providers showed that many are growing rapidly and enjoying growing donor bases and increasing support from local government and foundations. This growth indicates that the

¹⁵ For a state-by-state breakdown of funding, see Pula, Kevin, Douglas Shinkle, and Jaime Rall. "On Track: How States Fund and Support Public Transportation." Washington, DC: The National Conference of State Legislatures, 2015.

¹⁶ No research could be found that supports or refutes this anecdotal observation. The current Arizona State Plan on Aging, produced by the Division of Aging and Adult Services, provides no statistics with regard to the personal support networks of older adults. Research in this area could provide valuable information when forecasting how the level of demand for support services such as transportation will compare to other states.

services being provided are likely relevant and appropriate to the communities being served.¹⁷ Alongside this local engagement, significant isolation was found between peer agencies. Among some of the small providers there was a sense that there was no other provider like them or that there were not viable opportunities to coordinate with neighboring or kindred organizations. The result is effectiveness that is limited to each agency's local jurisdiction, an absence of tools for developing a transit network that connects communities, and too few opportunities for sharing best practices or developing a professional support network.¹⁸

This combination of local engagement and regional isolation is not unique to Arizona. In fact, it is one of the systematic challenges in transit that mobility management was designed to address. In response to these conditions, the Arizona Department of Transportation (ADOT) has embraced mobility management concepts and adopted a region-based approach in which ADOT directs federal funds to Arizona's regional planning organizations, which in turn hire mobility managers. The mobility managers work collaboratively with transit providers, human services agencies, healthcare providers, and other stakeholders to improve mobility within their region. They also meet periodically with ADOT and their peers to share knowledge and discuss topics of statewide importance. Interviews revealed that the work of the regional mobility managers enjoys positive regard and has resulted in improved regional connectivity and support for small providers.

With respect to agency use of information technology, a wide variety of types and levels of use was found, ranging from generalized software such as Microsoft Excel to complex agency-wide implementations of specialized software for scheduling and dispatch. Agency size was a primary factor in determining the type of software being used, as fleets of greater than ten or so vehicles are difficult to manage without a specialized tracking system. Among the agencies contacted, for example, Verde Valley Caregivers Coalition is in the later stages of implementing a specialized system with a high level of success. The key factors that appeared to lead to success among the sample was the simplicity of the solution, how standard the agency needs were, and the level of organization capacity and leadership the agency had to carry out a more complex implementation.¹⁹

Other States

Pennsylvania

Of the states evaluated through expert interviews, Pennsylvania funds rural transportation at the highest level by a significant margin. Since the 1980s, Pennsylvania has been a funder of demand responsive transportation through state lottery funds. Called the Senior Shared Ride Program, its current funding level is approximately \$66 million and serves as a backbone for transit throughout the state, resulting in a coordinated network of human service transportation service providers. Providers are municipalities, counties, transit authorities, or area agencies on

¹⁷ Appendix B, "Services Are Rooted in Community Needs."

¹⁸ Appendix B, "Significant Isolation for Some Providers. The Smaller the Provider, the Greater the Isolation."

¹⁹ Appendix B, "Some Agencies Have Found Effective Technology Solutions." See also recommendation 1.1 for how these instances of success can be replicated to other agencies.

aging. While anyone can use the service for a fee, for older adults, 85% of the cost is subsidized by the state, and in some areas even the remaining 15% cost is covered by the local area agency on aging. The same network also provides services for a state-funded persons with disabilities (PWD) program and Medicaid-funded non-emergency medical transportation. In addition, many fixed route transit providers also use the same infrastructure to provide their required ADA paratransit services. One impact of this high level of funding for demand-responsive transportation across Pennsylvania is that federal rural transit funds (Section 5311) are shifted toward rural fixed route, resulting in a higher level of rural fixed route services compared to most other states.

The PennDOT Senior Shared Ride program pays providers on a per-trip basis. This fact has had a strong influence on decisions about the kind of supporting technology deployed in the state. In the early years of the program, a variety of systems were being procured and deployed across the state, ranging from sophisticated scheduling and dispatch solutions to basic spreadsheets and even pen and paper. As PennDOT conducted its compliance audits of providers, they found significant inconsistencies in the reporting between providers, even between providers who were using identical software but had configured it differently.

About 2010, in response to concerns from the state Auditor General about the ability to have verifiable data to support per-trip reimbursements, together with a desire within PennDOT to have a more efficient method to procure technology systems that mirrored statewide contracts for vehicle purchasing, PennDOT opted to procure a single system and provide it directly to the vast majority of providers (42 of the 44, with only the two largest providers outside the system). In April 2012 Ecolane²⁰ was deployed to the first providers. The system was rolled out to providers at a pace of roughly 12 agencies per year.

With a single scheduling and dispatch system across most of Pennsylvania, PennDOT saw an opportunity to create digital access points to that system for riders through a “one-click/one-click” project. The initial primary focus of Pennsylvania’s one-call/one-click implementation was an array of tablet-based kiosks at “trip generators” such as VA medical centers and major hospitals. The kiosk system that was created worked well from a technical standpoint, but utilization was low, and the kiosks were eventually removed. Focus shifted to a single online tool, the open source one-click system from Cambridge Systematics, which was branded as Find My Ride.²¹ The one-click system serves as a service discovery tool for shared mobility options, including Uber, and is also the booking interface for the Senior Shared Ride service. Find My Ride is currently used in 12 of the 42 agencies using Ecolane and is most heavily used by Central Pennsylvania Transportation Authority (rabbittransit) where it was first deployed and has been most heavily promoted. According to John Taylor, Chief of Specialized Transportation Division at Pennsylvania DOT, online booking through one-click at rabbittransit accounts for about 6% of all reservations.²²

²⁰ www.ecolane.com

²¹ findmyridepa.com

²² John Taylor, Chief of Specialized Transportation Division, Pennsylvania DOT, Interviewed November 12, 2019.

Vermont

Vermont has the highest rate of volunteer-provided trips of any state. Forty-three percent of demand-responsive trips are performed by volunteer drivers. For 20 years, the entire transit budget of the state was strictly demand-responsive services. Then in 2007, the Vermont Agency of Transportation (VTrans) started supporting fixed route services. There are now dozens of rural fixed route lines across in the state.

For technology to support transit at the state level, VTrans created the Go Vermont website²³ in 2007 to bring together a range of commuter options along with carpooling, van pooling, 511 integration and a supporting call center. In 2012, VTrans piloted the production and maintenance of standardized data for all fixed route transit services around the state. They found the effort successful in making the state's transit investments easier for the public to use and administratively sustainable.

After making the investments into putting fixed route service data into a standard format (the General Transit Feed Specification, or GTFS²⁴), VTrans found that a high number of people were too far away from bus stops to use existing trip planners. In response, in 2014 VTrans started testing out describing general public dial-a-ride services in a computer-readable way using a recently developed standard, GTFS-flex²⁵. This work proved promising as a way to increase the visibility of demand-responsive services. Plans were put into place to work gradually toward producing GTFS-flex data statewide and adding a matching trip planner.

These plans were accelerated when the Federal Transit Administration awarded VTrans a grant to develop new functionality for the open source OpenTripPlanner (OTP) system.²⁶ With OTP already capable of generating trip plans for fixed route transit based on GTFS data, this project added the ability for OTP to also consume GTFS-flex data and include demand-responsive services in trip plans.

The new trip planner has now been added to Go Vermont's array of resources²⁷ at a time when transit enjoys a great deal of public support in Vermont. State funding for transit has increased 74% over the last 12 years. According to Ross MacDonald, the Public Transit Planner and Coordinator at VTrans, the biggest challenge now for public transportation in Vermont is managing the overwhelming support for transit. He states that a dominant question that comes from the legislature and voters is, "why can't you do more?"²⁸

Oregon

Despite providing a relatively small amount of funding to transit across the state, in 2011 the Oregon Department of Transportation (ODOT) became an active participant in generating

²³ www.connectingcommuters.org

²⁴ en.wikipedia.org/wiki/General_Transit_Feed_Specification

²⁵ github.com/MobilityData/gtfs-flex

²⁶ www.transit.dot.gov/sites/fta.dot.gov/files/FTA_MOD_Project_Description_-_VTrans.pdf

²⁷ plan.vermont.gov

²⁸ Ross MacDonald, Public Transit Planner and Coordinator, Vermont Agency of Transportation, Interviewed October 18, 2019.

computer-readable data about the state's transit services. ODOT assumed responsibility for generating GTFS data for all state-funded providers of fixed route transit that were not producing it themselves²⁹ and made the data available through a single website.³⁰

Six years later the fiscal picture changed significantly when in 2017 the Oregon Legislature created the Statewide Transportation Improvement Fund (STIF) through a permanent statewide payroll tax of 0.01%. This resulted in a significant funding increase for rural transit starting with disbursements in May 2019. Total forecasted STIF funding for the first 10 quarters, from Q3 of FY 2019 through FY 2020-2021, is \$253 million.³¹ With this sudden increase in funding, many rural agencies started exploring technology investments or made STIF grant applications for technology. In an effort to support smaller transit agencies that lacked the resources to evaluate which technologies are appropriate and when, in March 2019 ODOT launched a one-year transit technology pilot in which selected agencies could receive an ODOT-funded technology assessment carried out by contracted consultants.³² Seven agencies across nine Oregon counties were assessed. Though the pilot has only recently completed, and a full analysis has yet to be completed, anecdotal reports from agencies has been positive. One general manager reported to ODOT's Public Transportation Advisory Committee that their agency's assessment was useful for understanding the level of effort required to implement different technologies and communicating that information to the agency's board of directors.³³

ODOT is currently exploring continuing the pilot's work through a longer-term project that would extend to all interested transit agencies in the state. It has also started collecting data from providers throughout the state about their demand-responsive services as "a first step toward the development of GTFS-flex-based products and software."³⁴

Conclusions

A common thread in these three cases is that the application of technology is primarily driven by state-level investment strategies and funding mechanisms. Pennsylvania's per-trip funding of transit trips has resulted in a rigorous and centralized system that reaches down into the operational systems of nearly all the state's transit providers. In Vermont and Oregon, where formula-based per-agency funding requires less rigorous oversight, state strategies have focused on creating high-quality data that supports both the services of the individual agencies and statewide transit connectivity (since a complete set of GTFS data in the state allows riders to plan trips that traverse services through apps such as Google Maps). In the case of Oregon, providing direct technology support for small providers came only with a sudden rise in

²⁹ "Oregon-GTFS.com lists data for 31 Oregon transit services", trilliumtransit.com/2011/11/21/oregon-gtfs-com-lists-data-for-31-oregon-transit-services.

³⁰ www.oregon-gtfs.com

³¹ Oregon Department of Transportation Statewide Transportation Improvement Fund Legislative Report, February 2020, www.oregon.gov/ODOT/RPTD/Pages/STIF.aspx.

³² Full Path Transit Technology was the lead consultant for the project, together with Trillium Solutions.

³³ Meeting minutes from the November 2019 Public Transportation Advisory Committee, [www.oregon.gov/odot/RPTD/RPTD Committee Meeting Documents/PTAC_04Nov2019_MeetingMinutes.pdf](http://www.oregon.gov/odot/RPTD/RPTD%20Committee%20Meeting%20Documents/PTAC_04Nov2019_MeetingMinutes.pdf).

³⁴ "ODOT Invests in GTFS-flex," content.govdelivery.com/accounts/ORDOT/bulletins/248330d.

resources for those agencies. In all three cases, significant technology investment came only with a state-level plan and large or growing state funding of the transit services themselves.

The lack of a clear state-level strategy for rural transit technology,³⁵ combined with very limited state funding for rural transit means that there is very little opportunity for state-level solutions that would improve the usefulness of transit for all of rural Arizona. This limited organizational capacity stands in contrast to states like Pennsylvania, Vermont, and Oregon that have been able to use technology to create a network of services and systems that riders can learn about from a single state-supported source.

³⁵ Reports that involving state-wide technology where rural transit could be considered include the Statewide Strategic IT Plan from the Arizona Strategic Enterprise Technology Office (aset.az.gov/strategic-plan) and the ADOT Transportation Systems Management and Operations Plan (azdot.gov/sites/default/files/media/2020/02/ADOT-TSMO-strategic-plan.pdf)

Recommendations

These recommendations are designed to support the Rural Transportation Incubator's goals to provide technology that will help communities with their transportation programs. The recommendations respond to the challenges described in the findings and offer solutions that take into account the strengths and capacities currently found in the state and at individual agencies. They are influenced by current technology trends and recent developments in the transit technology field, while taking a perspective rooted in mobility management's focus on customer needs and coordination across jurisdictions. The recommendations are aimed at reducing the risks that come with implementing technology.

Recommendations are organized into three tracks:

1. **Agency operations**, with a focus on supporting agencies in the details of how they carry out their day-to-day work and meet regulatory requirements.
2. **Tools for discovering services**, with a set of proposed state-wide actions that support developing a common understanding of mobility options across rural Arizona, focusing first on acquiring and maintaining quality data.
3. **Regional coordination**, with a focus on mobility managers as providers and supporters of technology tools that draw in all stakeholders, foster collaboration, and build community.

Track 1: Operations

1.1 Establish a Statewide Technology Resource Center for Small and Rural Transportation Providers

Implementing specialized technology to solve complex problems always carries some level of risk. Those risks can be significantly reduced by providing targeted technical assistance.

Small and rural transportation providers rarely have their own IT staff. This puts them at a disadvantage when it comes implementing specialized technology. For smaller agencies, engaging technology is often burdensome at every stage. Even procurements that are perfectly executed from a procedural standpoint nonetheless often result in buyer's remorse. To address these challenges, this report recommends setting up a technology resource center (TRC) to give agencies access to expertise at strategic points:

- Provide outreach and education to inform agencies about how different specialized tools work and dispel common misconceptions about technology (e.g., believing in technology as a cure-all, underestimating the resources needed to implement and maintain a solution, uncritical acceptance of vendor promises, etc.).
- Carry out assessments of agencies to determine the areas where technology can best address pain points and constraints on growth. This could be done in combination with an evaluation of an organization's capacities to implement and maintain new systems. If an agency is interested in a particular type of technology solution or a specific vendor offering, the evaluation could take the form of a gap analysis combined with an analysis of the solution's likely return on investment (ROI). The gap analysis would identify what additional capacities the agency would need to have in place to succeed, while the ROI analysis would offer a framework for determining whether the investment in technology compares favorably with other possible investments that also advance the agency's mission.



Figure 3 — The key elements of a technology resource center

- Assist regions and agencies in understanding, updating, and following statewide procurement guidelines such as ADOT's intelligent transportation systems (ITS) architecture plan.³⁶
- Support agencies in gathering system requirements and crafting language for requests for proposals (RFPs). For those agencies opting to procure a technology solution, support with crafting of RFP language can assure that the agency effectively communicates the nature of the problem they are trying to solve, while avoiding overly vague or prescriptive requirements that end up being a barrier to successful vendor engagement.
- Conduct market research for specialized technologies that are of interest to multiple agencies in the state. A particularly beneficial first focus may be on tools supporting volunteer driver programs, though research could be done in areas such as in-vehicles cameras or GPS systems. If done in coordination with agency assessments, a range of concrete agency requirements and organizational capacities could be compared to the current solutions available in the vendor marketplace.
- Discover opportunities for shared solutions. With the completion of assessments of multiple agencies, it may be possible to identify opportunities where agencies can join forces in one or more aspects of technology procurement or operations. Such joint efforts could result in benefits such as reduced cost, improved coordination of services, or reduced duplication of effort. Where joint operations are being explored, assistance could be provided in developing a concept of operations and systems engineering approach for carrying the project forward.
- Provide implementation support. In some cases, deployment of technology can involve complex considerations that may outstrip an agency's capacity and should not be left entirely to the vendor to undertake. In such situations, targeted support or limited project management can mean the difference between success or failure when putting technology into place.
- Assist with developing maintenance plans and preparing for a system's eventual replacement. Once in place, ongoing maintenance of a system must be brought into the workings of an organization through policies and procedures in order for the technology to achieve its full potential lifespan. Once a system nears the end of its useful life, planning for its replacement can help minimize disruption to the agency.
- Evaluate past efforts. Retrospectively assessing technology projects can bring institutional learning to improve the outcomes of future efforts.

Providing support with targeted resources at these points can be summarized as setting up small community transportation providers for success. While no strategy can eliminate risk entirely when it comes to applying technology, supporting agencies in these areas can make the risks taken more conscious, strategic, and manageable. It can help make the successes observed in some agencies³⁷ the norm across the entire state, rather than a function of luck or local resources.

The resource center could also serve as an intermediary between small agencies and statewide plans which may be difficult for smaller entities to navigate, such as ADOT's ITS architecture. In

³⁶ azdot.gov/planning/transportation-studies/arizona-statewide-its-architecture

³⁷ Appendix B, "Some Agencies Have Found Effective Technology Solutions."

addition to the education efforts described above, the TRC may be able to work with ADOT and regional governments to help assure that state plans stay aligned with the needs of smaller transportation providers. It may also be able to help ADOT develop methods for streamlining the process for adhering to plans.

This recommendation is modelled after the approach currently being taken in the state of Oregon, where a recently enacted state-wide payroll tax for transit has led to new resources for rural agencies and with it an increase in desire for technology procurements.³⁸ Though still in its early phases, the program has resulted in better informed agency leaders and the likely avoidance of costly missteps.

With no new funding for rural mobility on the immediate horizon, putting such a program in place in Arizona would need to start small and work within the funds available from federal 5307 and 5310 grants. Initially, this could be carried out by assigning additional responsibilities to one of the planning organizations currently receiving funding for mobility management. There is precedent in Arizona for creating a such a state-wide resource leveraging existing regional infrastructure. It is important to note that any funding used to support this strategy would decrease the funding available for capital replacements, resulting in the need to reject requests from agencies for vehicles. For more than five years, ADOT has awarded the Southeastern Arizona Governments Organization (SEAGO) a grant to carry out driver training across the entire state. Unfortunately, the funding has decreased each year. While a similar model could be applied with respect to technology support, additional funding and partnerships would be needed. Over the longer term, a partnership with an academic institution may be possible, such as North Carolina has done with the Institute for Transportation Research and Education at North Carolina State University.³⁹

Once created, such a center may be able to get support from federally funded programs. The Federal Transit Administration funds technical assistance centers, all of which are increasing their level of support with regard to technology. These are the National Center for Mobility Management (NCMM), the National Rural Transportation Assistance Program (RTAP), and the National Aging and Disability Transportation Center (NADTC).⁴⁰

Particularly of note, there is also a newly formed technical assistance center that will soon be operating in a one-year pilot: The National Center for Applied Transit Technology (N-CATT), whose mission is more squarely focused on this domain. N-CATT “delivers expert, focused technical assistance to transit agencies and organizations in rural areas and small cities to use or develop transit technologies and innovations that make services more cost-effective and efficient. N-CATT’s work supports FTA’s mission and focus on innovation by developing and supporting transit programs and services in rural and small-city America.”⁴¹

Implementing a model like this would significantly benefit from close coordination with the Arizona Department of Transportation, the councils of governments, the metropolitan planning organizations, the Rural Transportation Advocacy Council, and the Local Technical Assistance

³⁸ For more details, see ODOT’s January 2020 legislative report on the impact of the new funds at www.oregon.gov/ODOT/RPTD/Pages/STIF.aspx.

³⁹ itre.ncsu.edu/focus/transit

⁴⁰ www.transit.dot.gov/ccam/resources/technical-assistance-supporting-communities

⁴¹ Ibid.

Program. Lack of funding is a significant barrier in Arizona. It is imperative to ensure these stakeholders deem this strategy a high priority and are committed to securing funding before work begins on any possible implementation. A phased approach might help to move the idea forward and build support for more robust investments in the future.

Cost Analysis

Phase/Element	Minimum	Maximum	Comments
Technology Resource Center (TRC) Coordinator ⁴²	\$63,252	\$159,838	1 FTE
Travel	\$3,000	\$10,000	In-state and out-of-state conferences for TRC staff
Contracted consulting services	\$25,000	\$100,000	The TRC may opt to contract out some technical assistance functions. Presented as a wide range that will depend on what services are provided, and how much is done in-house.
Provider cohort coordination	\$31,626	\$79,919	0.5 FTE, possibly spread across multiple mobility managers
Coordination travel	\$3,000	\$6,000	Reimbursement for periodic in-person meetings
Total	\$125,878	\$355,757	

To put the budget in perspective, funding to this degree would mean the Arizona Department of Transportation would be unable to fund the replacement of approximately four vehicles a year. This needs to be taken into account and the possibility of prioritizing this strategy over vehicles would need to be carefully vetted.

1.2 Organize Providers into “Birds of a Feather” Cohorts

Stakeholder interviews revealed that some providers experience significant isolation from peers. This isolation can be addressed with facilitated coordination of similar types of providers across the state.

This effort would leverage the existing inventories of all providers and categorize the lists into cohorts based on size and type of service. The recommendation is that each cohort coordinate with the state's Intelligent Transportation System Architecture Plan to align its structure and priorities based on its unique needs, and that the level of outside support vary based on the resources of the member agencies. For instance, a cohort of small volunteer driver programs may need to receive more proactive outside staff support in order to effectively coordinate, while a cohort of small urban agencies may be able to build and maintain their community with internal staffing and only occasional help from the outside.

⁴² Staffing costs are based on the national salary range for a regional planner, which currently ranges from \$45,180 to \$114,170 according to the Bureau of Labor and Statistics (www.bls.gov/oes/current/oes193051.htm). Benefits are estimated at 40% of salary, resulting in a range from \$63,252 to \$159,838.

While the creation and support of these cohorts could be carried out through the already recommended technology resource center, this does not need to be the case, especially if the creation of the center is delayed. It could instead be undertaken through Arizona’s current mobility manager network, where different mobility managers are tasked with supporting different cohorts.

Ongoing communication within the cohorts could be facilitated by simple collaboration tools such as a listserv as suggested by NAIPTA’s development director⁴³, or with a more robust tool such as the Basecamp system put into place for mobility managers in New York state.⁴⁴ See recommendation 3.1 for more details.

1.3 Use Mature and Generalized Technologies to Test and Innovate

While applying specialized technology carries higher risks, adapting more generalized tools can allow agencies to get started now and focus on testing out different solutions.

For agencies that are ready to start applying technology to solve their problems but don’t have the resources to undertake the work of undergoing a systems engineering process, there are still options available using tools that aren’t specifically designed for rural transportation, as Table 2 illustrates.

	Example Solutions	Considerations
Requesting/Booking	Google Forms, SurveyMonkey	Cyber Security
Scheduling	Google Sheets	Cyber Security, Design
Dispatch	Zello, Voxer	Device Management
Vehicle Tracking	Samsara, Geotab	Device Management
Reporting	Google Sheets	Cyber Security, Design

Table 1 — Examples of services to support different aspects of transportation operations

All examples are software as a service (SaaS) solutions that scale easily as needs grow. SaaS services generally have pricing structures that make them inexpensive or free to test out. Some, such as Google’s G Suite for Business, offer a free tier to 501c3 nonprofits. Each possible solution has associated considerations that can affect an agencies success in using it. These considerations are discussed in greater detail in recommendation 1.4.

⁴³ Interview with Kate Morley, Development Director, Northern Arizona Intergovernmental Public Transportation Authority (NAIPTA), September 19, 2019.

⁴⁴ Interview with Dwight Mengel, Chief Transportation Planner, Tompkins County Dept of Social Services, October 1, 2019.

Requesting or Booking Rides

For most agencies, receiving ride requests is a time-consuming process that requires staff to field phone calls and record the trip details by hand. Despite this challenge, many providers may be resistant to giving this task because of the perceived or real limitations this would place in their ability to serve their clients as responsively as they do currently. In addition, many agencies may not be in a position to take on the costs associated with such a partnership.

For those who are willing to partner and do have funding to support this activity, they may find that many riders need this level of service and there are also many who could submit their requests through an online form. There is now an abundance of services that offer highly configurable and secure ways to develop forms. Google Forms, SurveyMonkey, and Cognito Forms are just a few. While information from the forms would still need to be hand-copied into another system for scheduling, such tools would be low-cost way to experiment with reducing time on the phone.

Scheduling

Shared spreadsheets such as Google Sheets, through G Suite for Business, or Excel, through Microsoft's Office 365 service, make it very easy to put data into a lightly structured format, securely share it with others, and experiment with different approaches to coordination and collaboration in the scheduling process. These tools open the door to coordination in ways that emails, faxes, and phone calls do not.

Dispatch

There are now a wide range of services available to support communicating with drivers and knowing where a vehicle is located. Zello, Voxer, and others provide app-based secure push-to-talk communication that also provides live location tracking. Agencies could start with standard smartphone devices and upgrade to more ruggedized equipment as needed.

Vehicle Tracking

This area covers low-cost equipment that can be installed in a vehicle to communicate its location, speed, and direction of travel. Many systems can also track unsafe driving activities such as hard braking and hard turning. Some can be connected to the vehicle's computer system and relay diagnostic information in real time. While some vendor systems are closed, others such as Samsara and Geotab provide interfaces for 3rd party systems to also gather vehicle location. This can be useful for asset management systems or fixed-route services that wish to make the vehicle's location available to riders.

Reporting

Collecting all the information needed to fulfill funder reporting requirements can be time-consuming and frustrating. Spreadsheet applications such as Excel have long been invaluable tools in this effort. Similar to the case of scheduling, a SaaS spreadsheet system may be able to make the job easier by making the process collaborative. In addition, agencies can have a measure of auditability through Google Sheets' ability to track who made changes to a document and when.

Using general purpose tools as described is not without risk. They may adapt poorly in the transition from testing to everyday use, or they may reach their practical limits quickly as an organization grows. Workarounds put into place to make the solution work may overshadow the original improvement desired. However, as tools that are more easily understood and can be managed in-house, they can keep agencies in the driver seat in a way that more sophisticated solutions may not, and they can allow for rapid prototyping and vetting of concepts at a relatively low cost. Building capacity as described in recommendation 1.4 can help reduce risks.

1.4 Build Capacity in Areas That Support a Range of Technologies

Developing capacity in these areas lays important groundwork for using a wide range of technologies.

Get Cyber Security Training

The recent explosion of web-based SaaS vendors and services offers immense possibilities for small agencies to expand capacity without needing to purchase hardware or install software. But because using SaaS solutions relies on moving agency data from on-premise computers to hosted vendor systems, the ways in which staff manage data security needs to evolve, especially if privacy laws such as HIPAA apply.

Cyber security training can address the risks associated with the trend towards SaaS solutions and data in the cloud. In addition to the great many trainings available online, it may be possible to partner with other local organizations that have already developed trainings, such as hospital systems.

Create Documentation

Good documentation is a highly underrated contributor to developing more effective organizational processes. Its value is broad and has benefits unrelated to technology, such as improved consistency of work and smoother training of staff or volunteers.

When it comes to technology, however, documenting can be truly invaluable because it makes a process explicit and conscious. These are key ingredients for successfully introducing a technology solution, since automation of work thrives when processes are clear, consistent, and agreed upon by all involved. For agencies ready to explore using specialized software, good documentation provides a vital head-start in thinking about the functions the software should provide.

Documentation also enables makes processes more available for evaluation. Once something is written down or presented on a whiteboard, a group can better examine it, ask questions, and tinker. “Is this the best way to do things?” “It looks like there’s a step here that’s a holdover from when we had different reporting requirements. Can we eliminate it?”

It can be challenging to prioritize the documentation of systems when there is so much other work to do and things seem to be working fine enough as-is. Also, documenting processes that have become rote or are carried out by only one person aren’t easy to get a grasp on. It may be

useful to have an outsider, a new hire, or a new volunteer do this work, since they come with a “beginner’s mind.”

Apply Design Thinking

This report has already identified that mismatches between the design of off-the-shelf specialized software and existing agency workflows are a challenge for small agencies.⁴⁵ In contrast, when using spreadsheets and other generalized tools as discussed in recommendation 1.3, it falls to agencies to develop their own document formats and design their own workflows. This shifting of power and responsibility from software vendors to agencies means that agencies can (within the limits of the generalized tools) create something themselves that is a very good fit for their needs. It also creates the potential for getting into trouble by creating systems that don’t work well over the long haul. Design thinking can help with this area of risk.

Design thinking is a methodology that helps structure the process for figuring out how to arrive at a good design, especially when the process being designed for is complex and the requirements are unclear. It defines five stages (empathize, define, ideate, prototype, and test) and acknowledges the nonlinear nature of design work by making clear that the development of a system can proceed through the stages in many different orders depending on the situation.

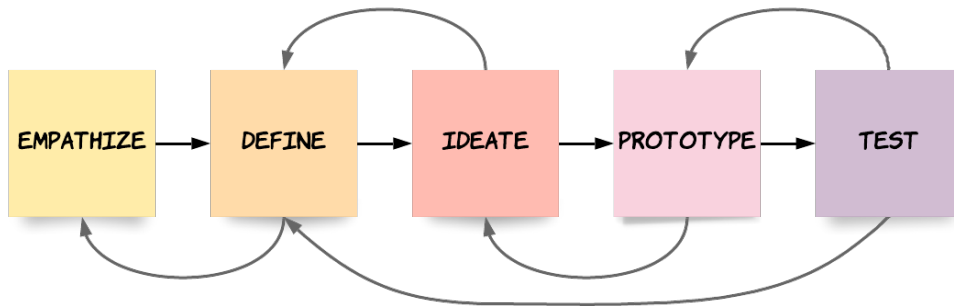


Figure 4 — The steps of design thinking. Progression through the stages varies widely with the situation.

The field of design thinking has been growing since the 1960s, and there is currently a wealth of literature available on the topic. For those wanting to learn more, a collection of introductory articles from the Interaction Design Foundation is a good place to start.⁴⁶

Build Systems to Manage Mobile Devices Effectively

Much of the interest in technology for transit and specialized transportation has hinged on using mobile devices — smartphones, tablets, and screenless devices plugged in vehicles — to enable real time tracking of activities in the field. The potential of these tools to transform operations is substantial but can be stymied if appropriate support systems are not in place. Additional challenges include potential resistance from agencies and an inability to allocate funding for this purpose. If these challenges can be overcome, key elements include:

- Hardware support and replacement plans that account for the higher likelihood of device breakage and failure in the relatively hostile environment of daily use in the field. Rural

⁴⁵ See “Problems Applying Technology”, page 4.

⁴⁶ www.interaction-design.org/literature/topics/design-thinking

communities often don't have easy access to specialized equipment, so an in-house inventory of spares may be needed.

- Operating system and software support plans to promptly update devices when security patches are released. Device or cellular service providers may offer service plans for this.
- Robust training regimens that account for the logistical challenges of working with staff that may be uncomfortable with the technology, rarely in the office, or difficult to bring together in one room. Plan for extra time and staffing.

Track 2: Tools for Finding Mobility Options

2.1 Create a System to Collect and Maintain Detailed Service Information

Without solid, detailed data on what mobility options exist in the state, providing robust tools to riders, planners, and policymakers is difficult. A recently improved data standard can help.

The work by Age Friendly Arizona to create a single directory of all the transportation services in the state reflects the importance of having a single understanding of specialized and rural transportation resources. This effort develops essential institutional capacity to collect and maintain information about services throughout Arizona. The directory is a significant improvement over forcing riders to hunt for services across many websites and multiple phone calls.

Until recently, directories such as the one created by Age Friendly Arizona were the only viable way to aggregate information about mobility options. The idea of putting service information for the full spectrum of mobility options into a reusable format was not achievable, as no data standard existed. Fixed route transit providers have had the General Transit Feed Specification (GTFS) available as a way to create and broadly share information about their services since 2005⁴⁷, but flexible modes such as volunteer driver and human services programs, medical transportation, general public dial-a-ride, and deviated fixed route services have had no such tool. The result is that for riders, their families, and their support people, it can be hard to discover the best options and figure out how to access them at the level of a specific journey. When planning a particular trip, it can be confusing to figure out which services are available, what the costs are, and whether public transit is a viable option. For planners and policymakers, it can be daunting to think about all the services together, identify gaps, or connect services for desired social outcomes.

With the recent development of the GTFS-flex extension to the existing GTFS standard, demand responsive and deviated fixed route services can now also be described and presented together with fixed route for the purposes of rider trip planning, service planning, research, and analysis. This recommendation is to take advantage of this standard, build on the institutional

⁴⁷ en.wikipedia.org/wiki/General_Transit_Feed_Specification

capacity to collect and maintain service information that has been developed with the current directory, translate all available services into the GTFS-flex format (or, in the case of small urban fixed route services, GTFS), and put systems into place to update the feeds as services evolve. This builds the necessary foundation for the two recommendations that follow on this track and which make an investment in GTFS-flex worthwhile.

As a well-documented format that can be created and edited by hand,⁴⁸ the work of putting service information into the GTFS-flex format could be carried out by in-state resources. However, the work does involve developing specialized expertise, and it may be more cost-effective to contract this work out to third parties. Such an approach would allow the focus of in-state resources to be on tracking the services across the state so that the data stays current.

2.2 Plan and Implement a State-Wide “One-Click” System

Use the power of GTFS-flex data to create a public-facing web-based system that serves as a single point of entry for trip planning and includes the full range of services available in rural Arizona.

With a system in place to gather and maintain detailed service data, the state is on solid ground technically to implement the “one-click” side of a “one-call/one-click” system. This is comprised of a web-based application that provides a single authoritative source of information that will allow riders, information and referral specialists, providers of medical and social services, and mobility managers to have access to comprehensive and complete information on transit, paratransit, and other personal mobility services. It includes a trip planner which, when supplied with data in the GTFS and GTFS-flex formats, has the ability to create detailed itineraries that include both fixed route and demand-responsive transportation services.

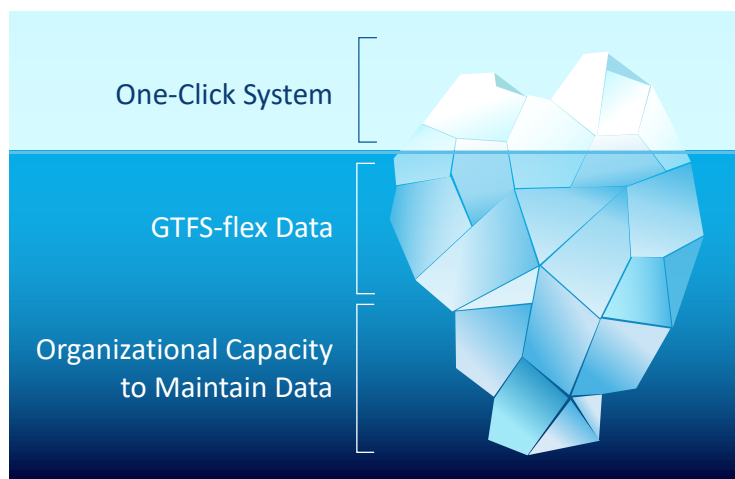


Figure 5 — A public-facing one-click system relies on good data and strong systems to keep it accurate over time.

One-click systems were first developed in 2011 as part of the Federal Transit Administration’s Veterans Transportation and Community Living Initiative (VTCLI), a \$60 million grant program to support technology and coordination in the human services transportation sector. The first system was developed as open source software⁴⁹, and proprietary solutions with a similar set of features have emerged since then. While the initial software development of the open

⁴⁸ At their core the GTFS and GTFS-flex formats are a collection of specially named and formatted spreadsheet files assembled into a single compressed “ZIP” file.

⁴⁹ github.com/camsys/oneclick-core

source platform was a multi-million-dollar effort, the current software costs for an “off-the-shelf” one-click implementation are now a fraction of that original investment.

Because the original one-click system predates the development of GTFS-flex, it relies on its own non-standard storage format for tracking demand-responsive services, and so cannot provide the reusability benefits described in recommendation 2.1. However, an open source trip planner that is capable of consuming GTFS-flex has recently been developed⁵⁰ and is ready for integration into a new version of the one-click platform, and while multiple agencies around the country are actively exploring making investments to carry out this work, software development has not yet begun. This leaves Arizona with three options to deploy a one-click system while also leveraging the benefits of the GTFS-flex standard:

1. Delay implementation entirely until another entity eventually funds the necessary work.
2. Implement a system now using the original one-click version (without the ability to feed it GTFS-flex data), and then upgrade when another entity funds the necessary work, which is likely to be completed in Q3 2021.
3. Pay for the necessary development work, resulting in no need to wait.

Based on Arizona’s current institutional capacity, this plan recommends option 1, where the delay is used as a period to develop the necessary shared vision needed to make a one-click system successful. In this scenario, creation of GTFS-flex feeds could be timed to coincide with the implementation of the one-click system. The costs for all three options are described below.

Creating a successful one-click system also involves key non-technical elements, including coordination to collect and maintain the data, a marketing plan, a governance structure, and establishing ways to measure the system’s success. Many of these efforts are already under way as part of the directory of services developed by the Age Friendly Arizona initiative. Advancing these non-technical elements relies on institutional capacity building, which would be supported by putting a technology resource center in place, as described in recommendation 1.1. For an in-depth examination of these non-technical elements, see *NCHRP Report 832: State DOTs Connecting Users and Rides for Specialized Transportation*, which is referenced in the literature review portion of this report.

⁵⁰ MacDonald, Ross, “Mobility on Demand (MOD) Sandbox: Vermont Agency of Transportation (VTrans) Flexible Trip Planner”, FTA Research Report No. 0150, Federal Transit Administration, 2020.

Cost Analysis

The following cost analysis is based on estimated level of effort to implement a one-click system and maintain it for one year. It includes the costs for collecting and maintaining GTFS-feeds as described in recommendation 2.1. Minimum total cost assumes option 1 described above, while maximum total cost includes the maximum cost for option 3.

Phase/Element	Minimum	Maximum	Comments
GTFS-flex agency outreach and project management ⁵¹	\$15,813	\$39,960	0.25 FTE
Initial GTFS and GTFS-flex creation	\$24,000	\$48,000	30 - 60 feeds @ \$800 average cost
Annual GTFS & GTFS-flex update costs	\$12,000	\$24,000	30 - 60 feeds @ \$400 average cost
One-click base deployment	\$25,000	\$35,000	0.5 FTE, possibly spread across multiple mobility managers
One-click annual hosting and support	\$18,000	\$35,000	Depends on level of proactive support and upgrades
Option 1: Delay one-click deployment until "2.0" is complete.	\$0	\$0	Cost avoidance by waiting until another agency pays for this development (likely complete Q3 2021)
Option 2: Upgrade to "one-click 2.0" with ability to ingest GTFS-flex feeds	\$10,000	\$20,000	This cost would apply in the case of deploying "one-click 1.0" early, then upgrading to "one-click 2.0" after another agency has developed it.
Option 3: Develop "one-click 2.0" with ability to ingest GTFS-flex feeds	\$100,000	\$150,000	This cost would apply in the case of assuming the full cost of developing GTFS-flex functionality
Initial project management, 1 year	\$31,626	\$79,919	0.5 FTE
Total	\$126,439	\$411,879	

It should be noted that the availability of funding needs to be thoroughly explored and confirmed, particularly for the long-term.

⁵¹ Staffing costs are based on the national salary range for a regional planner, which currently ranges from \$45,180 to \$114,170 according to the Bureau of Labor and Statistics (www.bls.gov/oes/current/oes193051.htm). Benefits are estimated at 40% of salary, resulting in a range from \$63,252 to \$159,838.

2.3 Use a “One-Click” System to Encourage Coordination Across All Modes

The inefficiency of demand-responsive transportation modes mean that they likely will not be capable of scaling up to the level necessary to meet growing demand. Thoughtful service design can connect less efficient modes with more efficient, higher capacity ones. Technology can help the public understand the design.

As demand grows for shared mobility options, it will be important to create a network of services that maximize efficiency wherever possible. Achieving such a network is primarily an exercise in service design: building out the appropriate routes, stops, schedules, and transfer points that respond to the peculiarities of the area being serviced. While service design needs to take the lead, technology can assist in presenting to the general public a clear picture of how the services combine and result in a coherent network of services that cover a range of needs and abilities, from volunteer driver programs to inner- and inter-city fixed route transit. Once in place, the one-click platform can work with Arizona’s existing regional mobility manager infrastructure to present these localized views of public transportation options.

Two regions in Oregon provide examples of a direction which could be applied through a state level website. The Northwest Oregon Transit Alliance (NWOTA) developed a plan to coordinate services across northwest Oregon.⁵² The plan included a number of key elements, including a governance structure and the formation of a route coordination subcommittee. With respect to technology, it includes the development of a single website that presents its five member agencies as a single unified service.⁵³ The site includes a trip planner leveraging Google’s transit planning functionality, which in turn relies on GTFS data produced by ODOT. NWOTA has found the approach to be successful.

⁵² www.nworegontransit.org/wp-content/uploads/2018/10/Connector-Management-Plan_FINAL-10-21-16.pdf

⁵³ www.nworegontransit.org

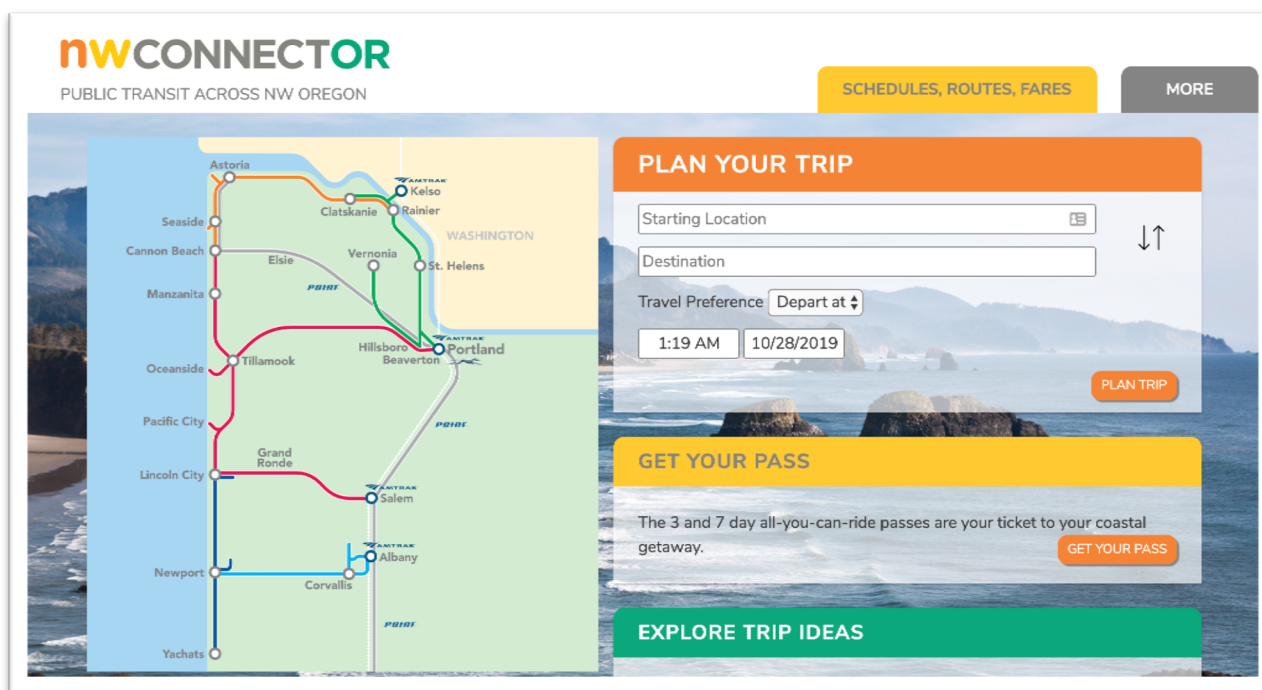


Figure 6 — A screenshot of the Northwest Oregon Transit Association website, branded as the NWConnector.

In 2019 another Oregon region, the Columbia River Gorge, replicated the approach with its own website that is based on the same underlying software code.⁵⁴ The Gorge website takes the additional step of highlighting specialized transportation options alongside fixed route options. The services for older adults and people with disabilities in highly rural Sherman County are presented prominently in the home page's service map and in the listing of network agencies, as shown in Figure 7.

⁵⁴ gorgetranslink.com



Figure 7 — A screenshot of the Columbia Gorge Translink website. Sherman County's demand-responsive services appear in red on the right side of the system map.

A website that include all modes meets two objectives: it serves members of the general public with useful tools tailored to them and stimulates the both transit and specialized transportation providers in the region to work together to create a unified “mobility for all” vision. The latter objective further addresses the isolation which is noted in the System Evaluation and partially addressed in Track 1 recommendations.

2.4 Use GTFS-flex Data for Stakeholder Engagement

Granular data is powerful. Foster a data and technology tool ecosystem that can serve to highlight the importance of rural public transportation.

Alongside the creation of a one-click system for Arizona, this plan recommends using the GTFS and GTFS-flex datasets as additional tools to engage existing and new stakeholders to better understand and communicate the role of mobility in the health and welfare of people living in rural Arizona. The data can be used by planners and policymakers to create a single statewide view of mobility, overlay need with current service to identify gaps, and aid in planning.

While there is not yet a robust set of analysis tools capable of consuming GTFS-flex data, examples from fixed route transit provide a glimpse into possible directions for the development of an ecosystem. The majority of the following tools have been developed just within the last five years:

- In Oregon, the Oregon Department of Transportation (ODOT) publishes all Oregon GTFS feeds it has access to in one central location,⁵⁵ making it easy for software developers, researchers, and engaged citizens to build tools from the data. The substantial state investment for transit offers the flexibility needed to implement this approach.
- ODOT has also developed TNeXT, an open-source tool for analyzing transit systems and state transit connectivity.⁵⁶ The tool was originally developed at Oregon State University, and support is now managed by a private developer under contract with ODOT.
- Vendors such as Trillium Solutions, Remix, TransLoc, and Conveyal are private developers or software-as-a-service providers that produce GTFS feeds or provide analysis tools using GTFS data.⁵⁷
- Transitland⁵⁸ is an aggregator of publicly available GTFS feeds. In addition to its comprehensive directory of feeds, it provides a set of visualization tools.
- Mapnificent⁵⁹ is an open source and an easy-to-use interface for creating maps that show the service accessibility through visualizations of the travel time that various transit services around the world provide.

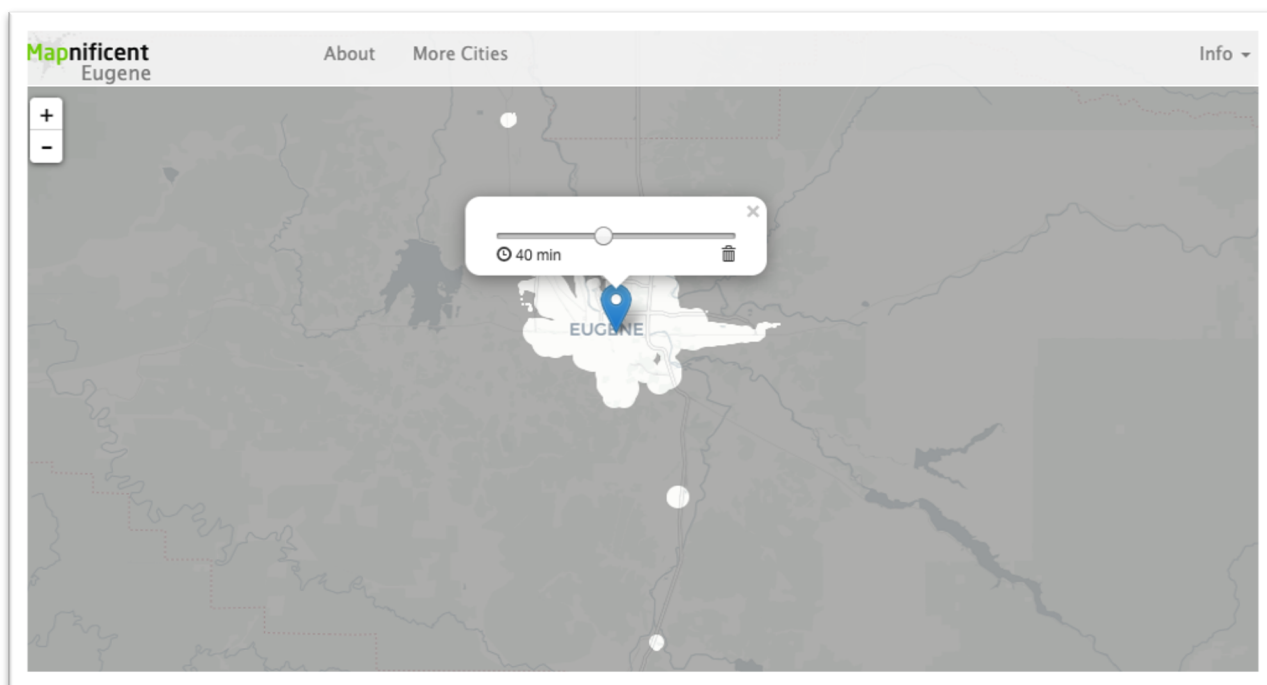


Figure 8 — Mapnificent map showing transit access from downtown Eugene Oregon in 40 minutes, using GTFS data. Similar tools could be created to include demand-responsive services using data in the GTFS-flex format.

⁵⁵ oregon-gtfs.com

⁵⁶ A live version of the app can be found at oregon.tnext.io. The software code is located at github.com/ODOT-PTS/TNeXT

⁵⁷ trilliumtransit.com, www.remix.com, transloc.com/gtfs-builder, www.conveyal.com/analysis

⁵⁸ transit.land

⁵⁹ www.mapnificent.net

What all these tools share is the ability to make the value of transit more readily understandable, either to the lay person or the planner. This is important in a society that is increasingly driven by data. The absence of equivalent tools for rural flexible mobility options has the effect of reinforcing the marginalization that rural mobility services experience. Currently, rural mobility is effectively invisible in many settings where mobility options are being evaluated because the tools to easily visualize and analyze it do not exist. The development of analytical tools for demand-responsive services, or the expansion of the specific tools named above so they also consume GTFS-flex, would open the door to new understandings about the existence and value of public and community-based transportation in rural mobility.

An early example of this has already occurred with the upgrading of an open source trip planner so that it can consume GTFS-flex data and provide trip plans that include general public dial-a-ride service for rural Vermonters.⁶⁰

The software development work required for these kinds of efforts is not particularly difficult, but it does require resources and, more importantly, champions acting in the interest of rural communities and advocating for developing a more sophisticated and compelling understanding of rural mobility. A technical resource aligned with the interests of rural transportation like the one described in recommendation 1.1 would assist in this effort.

⁶⁰ MacDonald, Ross, “Mobility on Demand (MOD) Sandbox: Vermont Agency of Transportation (VTrans) Flexible Trip Planner”, FTA Research Report No. 0150, Federal Transit Administration, 2020.

Track 3: Regional Coordination

3.1 Expand Use of Tools for Collaboration

Regional mobility managers should continue to take the lead in setting up and maintaining tools to connect transportation providers, stakeholders, and the general public.

Collaboration is foundational to the concept of mobility management. Technology can serve to expand the reach of coordination activities, especially as the distances involved, the number of participants, and the complexity of projects grow. Many software-as-a-service applications that are aimed at supporting teams within a business are also a good fit for the engagement work of mobility managers, though some adjustments may be needed. Table 3 lists example solutions.

	Example Solutions	Considerations
Conferencing	Zoom, Skype, Hangouts	• Cost increases as use grows • Proper configuration • Equity of access • User training and support • Online culture and facilitation
Group Email	Google Groups, Listservs	
Real Time Messaging	Slack, Chanty	
Project Management	Basecamp, Asana	

Table 2 — A listing of sample technology tools for coordination.

Many of the tools and services listed offer a tier of service with no fees. At minimum this allows for easy testing, but the tier may be generous enough for some ongoing uses. When using “free” tier services, care should be taken to factor in the eventual cost if there is the possibility of needing to upgrade as use grows.

Even the easiest to use and no-fee services have costs involving setting them up and making them available to users. For example, group email tools have a wealth of configuration options that can add extra polish and ease of use for the participants.

The power of each tool also comes with the risk of it being either difficult or impossible to use for some of the target audience. The issues can include inexperience with computers, disability, lack of internet access, or inability to afford the high bandwidth required. The concepts of progressive enhancement and, conversely, graceful degradation can be useful here. This is when the requirements for using a tool can vary depending the ability of the user. Examples include videoconferencing platforms that allow participants to call in by phone and project management tools that support participation by email as well as through a web application.

Mobility managers will need to balance the sophistication of a given tool with its ease of use. With a greater the number of features comes a greater likelihood of increased need for training to use the application effectively. If the aim is to include members of the general public, the simplest option that meets the core need is often the best one.

Like in-person events, on-line collaboration works best in an environment where all participants feel that the venue is a productive and safe place to contribute their perspectives. Online

settings can pose challenges to effective engagement due to unfamiliarity with the tools and difficulty communicating emotion and social cues. Because of this, the normal tools of facilitation, such as a clear agenda for a meeting, become even more important. In addition, a more hands-on approach can make the difference, through such strategies as normalizing the awkwardness people may feel and actively inviting the participation of people who are engaging less. The more public a venue becomes, the more helpful it can be to have documents that familiarized newcomers to the technology and things to expect in general. Many online communities have developed codes of conduct to clearly articulate expected behavior and what the response to inappropriate behavior will be.

Tools for Operational Coordination

The previous recommendations in this report have focused on coordination at the levels of technical support or service discovery. These are the areas most likely to yield the greatest benefits, either because they are lower-risk endeavors, or they actively reduce risk with regard to technology. Beyond these areas there is also the application of technology to coordinate daily operations between independent providers. Examples include joint scheduling, joint dispatching, or exchanging trips requests for purposes of saving money or improving rider experience.

Such operational coordination has been the target of numerous efforts across the country and is a frequently named goal in state and local coordinated human services transportation plans. It is this author's experience that this use of technology is difficult to get right in a reliable way, as success depends so highly on local conditions, the tools used, and the organizational capacities of all involved.

While there are documented cases of clear benefits from such coordination,⁶¹ they are unfortunately too rare. The success stories have depended on a high degree of planning and ongoing institutional support. For this reason, the recommendation is not for any specific operational coordination, but rather for further study at the regional level focused on evaluation of the expected benefits and organizational capacities of key players.

⁶¹ The coordination project between Denver RTD and Via Mobility in Longmont Colorado is one noteworthy case. It is described in the Mobility Services for All Americans Case Study Report, rosap.ntl.bts.gov/view/dot/36956.

Timeline

Each track is designed so it can begin immediately following the sequencing recommended below. Before this timeline can be activated, the realistic availability of ongoing funding needs to be confirmed.

0-5 Months

- Track 1:
 - Identify possible resources in the state to play the role of technology resource center.
 - Inventory all transportation providers to engage and form “birds of a feather” cohorts at a state level.
 - Begin cohorts with quarterly meetings (schedule 2 within the first 6 months).
- Track 2:
 - Inventory all agencies for which GTFS or GTFS-flex feeds need to be created.
 - Begin procurement process for a contractor to create required GTFS-flex feeds
 - Assemble stakeholders to decide the entity to deploy and manage the one-click system.
- Track 3:
 - Identify the highest value areas for technology-based collaboration tools.
 - Select the best candidate tools for each area.
 - For each candidate, develop estimates for initial and ongoing costs, assuming use is expanded broadly.
 - Select the most promising candidate tools, test, and begin initial use.
 - Develop initial documentation and codes of conduct, as needed.

6-11 Months

- Track 1:
 - Select the entity to house the technology resource center.
 - Develop a first-year plan, using the action areas described in the Track 1 recommendations, prioritized by input from participating stakeholders.
- Track 2:
 - Select contractor and begin GTFS and GTFS-flex feed creation.
 - Formalize selection of the entity to deploy and manage the one-click system.
- Track 3:
 - Expand use of successful technology-based collaboration tools
 - Further develop documentation and codes of conduct based on the feedback from agencies and public users.

12-23 Months

- Track 1:
 - Implement the first-year plan for the technology resource center.
- Track 2:
 - Carry out procurement for the one-click system.
 - Deploy the system with generated GTFS and GTFS-flex data.
- Track 3:
 - Review the state of the marketplace in each priority area for new and promising vendors or services.
 - Review use of tools in use to evaluate whether a new direction is warranted based on changes in needs or changes in the marketplace.

Conclusion

While many are pinning hopes on technology as a way to address the increasing demand for mobility options in rural areas, there is significant risk to bringing complex technologies into the equation. Technology in itself does not bring inherent value to rural transit or any other human endeavor. Rather, its value comes from being appropriately designed to fit the circumstances where it is being applied. Poorly applied, it can cloud issues or aggravate problems by draining resources from more effective solutions.

The recommendations in this report are aimed at reducing the risk that comes with implementing technology. For Track 1, the primary focus is on creating the conditions in the state where agencies will be able to get the support they need to address their particular challenges. A technical resource center, along with organizing agencies leaders into “birds of a feather” cohorts, can help equip agencies to bring appropriate technology into their organizations. For tracks 2 and 3, risk is addressed by recommending tools that have been used elsewhere and are mature enough to demonstrate that they are investments that can bring returns. Applications that present services to the public through trip planning have reached a level of maturity that make them a reasonably safe bet, and producing the data needed to feed these systems can have valuable spin-off benefits. Robust tools that allow users to gain insight into all their travel options can help bring together conventional transit and more specialized services into a single network that can grow with demand in a sustainable manner.

Recommendations have been split into areas that focus on agencies, the state as a whole, and regions, in an attempt to respond to the diversity of challenges that rural areas face in expanding the reach of mobility options. As one author reviewing the state of rural mobility technology put it, “there is no one problem, [so] there is no one solution”.⁶² While this can be a daunting reality, it also means that every stakeholder can pick an area of effort and work to improve the current conditions, regardless of their position or area of influence.

For all recommendations, there is the matter of who the best actors are to carry out the recommendations presented here, and how they should secure funding. While there are short-term opportunities to build on the current regional network within Arizona’s current federal Sections 5310 and 5307 allocation, this report does not focus on who should do the work long term. Rather, it strives to identify the areas of effort that are most likely to reap rewards. Organizations that play a coordinating role for rural mobility, such as Maricopa Association of Governments and the Arizona Transit Association, are encouraged to use this information to spur conversations and decisions at a state-wide level through an inclusive process that best fits Arizona’s needs.

⁶² Broderick, Andrew, “The Future of Rural Transportation and Mobility for Older Adults: Current Trends and Future Directions in Technology-enabled Solutions”, CITRIS, the Banatao Institute, and Grantmakers In Aging, April 2018.

Appendices

Appendix A: Literature Review

This literature review addresses research assessing the effectiveness of data platforms and technology applications used to improve the efficiency and effectiveness of transportation services for older adults. It covers five different topic areas where demand-responsive and specialized transportation have received significant attention through research: Computer-aided scheduling and dispatch, one-call/one-click and mobility as a service (MaaS) systems, coordination of operational technology systems, and transportation network companies.

Broadly speaking, there is a notable lack of research addressing technology for transit in the rural context. Our review has found that research on rural transportation gives short shrift to technology, while research on transit technology, even specialized transportation technology, leans heavily toward urban and suburban environments. For this reason, some of the literature reviewed was selected for its focus on specialized transportation in urban areas, as a surrogate for rural areas. While there are some commonalities, caution should be taken applying such research about urban specialized transportation to mobility in rural settings.

There was one exception to the absence of research in rural areas. Two reports from FTA Research, reports 0006 and 0065, focus on using technology for coordination in rural areas. Both these reports capably demonstrate the high barriers facing rural agencies that wish to carry out operational coordination.

However, no recent research could be found that addressed technology for general rural transportation operations, be it demand-response, fixed route, volunteer driver programs, or other modes. TCRP's "Guidebook for Selecting Appropriate Technology Systems for Small Urban and Rural Public Transportation Operators" was released in 2002, before the rise of software as a service and networked multipurpose mobile devices. The result is a system-level blind spot with regard to where technology today can bring sufficient value to be worth the effort required to put it into place and maintain it. It is precisely in that blind spot that small rural agencies are prone to believe sales pitches, act on unconscious cultural mythologies that hold out technology as a sure-fire benefit and assume that solutions which bring value in denser settings will also work in sparser ones.

Computer-Aided Scheduling and Dispatching Technology

David S. Kessler
TCRP Synthesis 57

Computer-Aided Scheduling and Dispatch in Demand-Responsive Transit Services
2004

onlinepubs.trb.org/onlinepubs/tcrp/tcrp_syn_57.pdf

Computer-aided scheduling and dispatching (CASD) systems were a topic of intense research in the 1980s and 1990s. Research, except for narrow scenarios, has waned in recent years. The referenced synthesis from 2004 collects the state of research up to that point. While dated, key elements of the report are unaffected by the passage of time:

- Application of route optimization requires consideration of trade-offs. "The more customers who are added to a vehicle (through effective use of CASD), the greater is

the efficiency, but also the greater are the passenger ride times. Thus, the quality of service ultimately is diminished.” (p. 9)

- “Schweiger and McGrane found that increased worker stress and lower job satisfaction often accompany the implementation of new technologies, including CASD systems. They found that operating employees have a wide variety of responsibilities and thus it is difficult for them to maintain a technical focus.” (p. 10) With a labor pool (particularly in the case of drivers) that is consistently near or past retirement age, use of technology is likely to continue to pose challenges to implementing CASD systems in rural settings for years to come.

This report and earlier ones like it show that some benefits to CASD focus their findings on urban and suburban areas. There appear to be no reports specifically assessing the value of CASD in a rural setting and comparing it to approaches that are smaller in scale, more manual in their operations, and emphasize relationship-based knowledge of the problem space. This gap means that efforts to apply large scale urban solutions to rural spaces should be evaluated with caution. Extra costs such as driver training and difficulties with network connectivity for vehicles that usually don't apply in an urban space should be clearly factored into cost-benefit analyses.

One-Call/One-Click & Mobility as a Service Systems

The below-referenced research focuses on rider-facing technology as a way of reducing the barriers to accessing specialized services.

Will Rodman, Dan Berez, Sarah Moser, and James Choe

NCHRP Report 832

State DOTs Connecting Specialized Transportation Users and Rides

Volume 1: Research Report

Volume 2: Toolkit for State DOTs and Others

July 2016

www.nap.edu/catalog/23506

www.nap.edu/catalog/23507

NCHRP Report 832 is the most relevant and in-depth research on this topic. It delves deeply into the results of a raft of deployments of “one-call/one-click” systems that took place in the early 2010s and provides a proposed evaluation framework for states and regions exploring the development of their own systems. The experiences of prior implementations demonstrate that institutional capacity is key to success in providing the necessary coordination for one-call/one-click system to present a coherent set of services. While the introduction of technology into coordination brings exciting new opportunities, but it also requires significant additional technical expertise and continued investment in order to stay relevant in a rapidly changing landscape.

Volume 2 of the report provides a toolkit with a proposed step-by-step method to create a successful one-click system:

- Identify geographic scope
- Select target users, trips, and modes
- Analyze existing information and referral “linkage” functionality
- Determining desired “linkage” functionality
- Developing evaluation criteria and an evaluation process
- Evaluate sources of funding
- Create a marketing plan

The level of effort described in this report underlines the importance of a strong actor to implement a successful system.

Jana Lynott

AARP Public Policy Institute

Universal Mobility as a Service: A Bold Vision for Harnessing the Opportunity of Disruption

September 2018

www.aarp.org/content/dam/aarp/ppi/2018/08/universal-mobility-as-a-service-aarp-ppi.pdf

This white paper reviews the state of the art for mobility as a service (MaaS) and applies an equity lens to it, resulting in a proposal for a “Universal MaaS” that applies universal design principles linked to AARP’s age-friendly communities initiatives. In the realm of technology, the report addresses the need for data standards such as those described in TCRP Report 210 (described below) and profiles a successful system in Denmark that builds on just such a standardized data exchange system. The Danish example provides a real-world example of scalable coordination rooted in streamlined financial transactions. This sort of coordination, largely unknown in the United States, could be an area of interest for philanthropic investment in Arizona.

Andrew Broderick

CITRIS, the Banatao Institute, and Grantmakers In Aging

The Future of Rural Transportation and Mobility for Older Adults: Current Trends and Future Directions in Technology-enabled Solutions

April 2018

www.giaging.org/documents/180424_CITRIS_rural_mobility_paper_F.pdf

This report provides an overview of hot technology topics in the transportation sector and applies them to the rural, older adult context. While he sees some future opportunity, the author makes clear that “The rural challenge [...] is the abundance of scarcity that exists, particularly when it comes to the availability of financial and technical resources, to support the adoption and implementation of technology solutions at scale.” He finds that “there is no one problem, [so] there is no one solution”, and because “each community has unique needs and resources to address needs”, he suggests “first leverag[ing] the resources, supports and infrastructures that already exist within communities before introducing a new solution to an old problem.” These findings echo this report’s focus on working with the specific needs of each agency when it comes to technology for operations.

Coordination of Operational Technology Systems

Anna M. Nalevanko, Andrew Henry, et al

TCRP Report 76

Guidebook for Selecting Appropriate Technology Systems for Small Urban and Rural Public Transportation Operators

2002

onlinepubs.trb.org/onlinepubs/tcrp/tcrp_rpt_76.pdf

Judith M. Espinosa, Matthew R. Baca, Amy D. Estelle, Nancy Bennett, Geri Knoebel, Mary E. White, and Josette P. Lucero

Client Referral, Ridership, and Financial Tracking [CRAFT] Transit Management System: CRAFTing a Bridge to Coordinated Interagency Transportation

November 2002

citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.129.5435

Jon E. Burkhardt, Charles A. Nelson, Gail Murray, and David Koffman
TCRP Report 101
Toolkit for Rural Community Coordinated Transportation Services
2004
www.nap.edu/catalog/13751

Patricia Monahan, et al
TCRP Report 105
Strategies to Increase Coordination of Transportation Services for the Transportation Disadvantaged
2004
www.nap.edu/catalog/13784

US Department of Transportation Intelligent Transportation Systems
ITS Applications for Coordinating and Improving Human Services Transportation: A Cross-Cutting Study
May 2006
rosap.nhtl.bts.gov/view/dot/3930

David Ripplinger and Del Peterson
ITS Transit Case Studies: Making a Case for Coordination of Community Transportation Services Using ITS
September 2005
www.ugpti.org/resources/reports/downloads/dp-171.pdf

Pamela Couch and Marcy Jaffe
FTA Research Report No. 0006
Evolution of Intelligent Transportation Systems for Mobility Management and Coordination Serving California's Rural Frontier
January 2012
www.transit.dot.gov/research-innovation/evolution-intelligent-transportation-systems-mobility-management-and

This report describes a failed effort started in 2004 to develop a physical and virtual mobility management center (MMC), resulting in a “one stop shop” for mobility options in Modoc County, California. Key takeaways from the report:

- Data standards matter. Many of the goals for providing trip options could not be achieved due to the lack of a format to describe how rural transit works. In the years since this report was released, the GTFS-flex format has been developed, addressing aspects of this barrier.
- Technology for the urban context cannot be readily transferred to the rural context with the assumption that there will simply be fewer trips. The long distances involved and sparse distribution of mobility resources call for a different set of tools.
- The goals of the project were overly ambitious by a wide margin. In a single project they attempted a system that would provide planning, booking, and payment for a range of options. Integrating these three options remains today an elusive goal even for large transit providers.

Lynnda Basshgam and Dana Luttrull

FTA Research Report No. 0065

Lower Savannah Aging, Disability & Transportation Resource Center: Regional Travel Management and Coordination Center (TMCC) Model and Demonstration Project

October 2014

www.transit.dot.gov/sites/fta.dot.gov/files/FTA_Report_No._0065.pdf

This report represents the only detailed account that could be found where sophisticated technology for coordination was applied in a rural context. The unvarnished presentation of the project's challenges and outcomes gives a likely representative sample of the kind of problems that can arise when seeking to overcome both technical and institutional barriers at the same time. The "lessons learned" section in particular should be required reading for any group undertaking a similar effort. It contains such unequivocal statements as "double the training budget", and "plan for future sustainability and identify all ongoing fees." The key takeaway for Arizona is that technology for operational coordination at this scale should be undertaken only when institutional relationships are strong, the gains are clear, and the technology is suitably mature.

Brendon Hemily

FTA Research Report No. 0117

Reference Manual for Planning and Design of a Travel Management Coordination Center (TMCC)

December 2017

www.transit.dot.gov/research-innovation/reference-manual-planning-and-design-travel-management-coordination-center-tmcc

This research provides a framework for understanding different ways service coordination can be achieved through technology at different steps in a trip's life cycle, steps such as customer discovery of options, trip scheduling, dispatching, and reporting. While it makes sense to be cautious when considering large technology-centric projects of this scale, there are definite circumstances where the benefits are real. For such cases, this report presents the range of possible coordination methodologies options through a set of matrices that provide an easy-to-grasp set of visual tools for engaging stakeholders in conversations about system design and project phasing. It's emphasis on systems engineering, echoing FTA Research Report 0065 above, also provides reasonable advice for the level of planning required for such endeavors.

Transportation Network Companies

In recent years some transit agencies have looked to transportation network companies (TNCs) as solutions to provide transit in low-density settings where fixed route transit is not viable. The following research presents case studies for this new and rapidly evolving space. Key takeaways for rural and small urban service:

- "Transit agencies must ensure that enough TNC drivers are available to meet demand for requested trips." (Hansen, et al)
- Curtis, et al only evaluated one agency (RabbitTransit in Pennsylvania) whose service area includes rural areas, indicating that TNCs may only rarely be useful outside of urban and suburban environments.

This research indicates that, despite the high level of attention that TNCs receive, they are unlikely to be a solution for rural and small urban environments because rarely can they provide a reliable supply of drivers and vehicles to meet demand.

Todd Hansen, Zachary Elgart, Kristi Miller, and Maarit Moran

Texas A&M Transportation Institute

Existing Transportation Network Companies Used as a Part of Basic Mobility: White Paper

November 2018

groups.tti.tamu.edu/transit-mobility/files/2019/04/TTI-Existing-TNCs-Used-as-a-Part-of-Basic-Mobility-White-Paper.pdf

Terra Curtis, Meg Merritt, Carmen Chen, David Perlmutter, Dan Berez, and Buffy Ellis

Partnerships Between Transit Agencies and Transportation Network Companies (TNCs)

October 2019

www.nap.edu/25576

Data Standardization and Interoperability

The lack of data standardization for mobility management has long stood as an obstacle to robust technology-based coordination. Past large-scale attempts to coordinate service discovery, service access, or operations have been severely hindered by the absence of tools allowing independent systems to exchange data between one another. A series of three reports, starting with TCRP Transit IDEA Project 50 in 2012, have attempted to address this barrier by proposing increasingly specific and actionable standards.

The most recent report, TCRP Report 210, recently released as a pre-publication draft, proposes a detailed proposed data standard for describing trip requests and associated workflows. The development of standards for trip booking is a key requirement for the “universal mobility as a service” approach advocated by Jana Lynott of AARP in her report cited above. If a project in Arizona were to undertake operational coordination, it could require in its concept of operation and technology procurement that the proposed data format described in this report be used for communication between systems. If implemented successfully, the result of having data standards for “plug and play” functionality would be a system that is more flexible, maintainable, and scalable as services grow over time.

One potential use of these data standards is for booking functionality in a statewide or regional one-click system. In this case, a user could search the one-click system for options for a particular journey, select the most desirable option, and then, in the same one-click system, click on a “book” button to request the service. Behind the scenes, the one-click system would send a booking message to the selected service using the standardized message format. If every provider in a region agreed to accept such booking messages in the same format, then creating the “one-stop shop” envisioned in FTA Research Report 0006 could be much more easily achieved.

The final report in this category, FTA Research Report 0150, describes a recent project to expand the capabilities of an open source trip planner so that it can provide trip plans using demand-responsive services based on data in the new GTFS-flex data format. The resulting software was then implemented as a trip planner for the entire state of Vermont, including rural dial-a-ride services.

Jon E. Burkhardt

TCRP Transit IDEA Project 50

Developing Regional Mobility Management Centers

October 2012

onlinepubs.trb.org/onlinepubs/idea/finalreports/transit/transit50.pdf

Suzanne M. O'Neill and Roger F. Teal

TCRP Web-Only Document 62

Standardizing Data for Mobility Management

November 2013

www.nap.edu/catalog/22449/standardizing-data-for-mobility-management

Roger Teal, Niels Larsen, David King, Candace Brakewood, Charlotte Frei, and David Chia

TCRP Report 210 (Pre-Publication Draft)

Development of Transactional Data Specification for Demand Responsive Transportation

September 2019

www.nap.edu/25619

Ross MacDonald

FTA Research Report No. 0150

***Mobility on Demand (MOD) Sandbox: Vermont Agency of Transportation (VTrans)
Flexible Trip Planner***

January 2020

www.transit.dot.gov/sites/fta.dot.gov/files/docs/research-innovation/146891/fta-report-no-0150.pdf

Appendix B: System Evaluation

This system evaluation for specialized transportation in the state of Arizona is organized around a SWOT analysis. Each strength, weakness, opportunity, and threat is composed of “why it matters”, “what we learned”, and “implications for technology” sections.

The primary basis for the “what we learned” sections is interviews with stakeholders throughout the state of Arizona, both by phone and in person during a visit to Arizona conducted in late September 2019.

The “why it matters” and “implications for technology” sections are based on the experience of the staff of Full Path Transit Technology.

System Strengths

Regional Mobility Manager Network Enjoys Positive Regard

Why It Matters

Successfully applying technology in and for small transportation providers works best when agencies are not operating in isolation. While larger agencies may be able to carry out their own technology projects using in-house requirements gathering, dedicated procurement staff, and project management resources, those that do not have direct access to such specialized expertise will be able to move further and faster with the support of outside help.

What We Learned

The Arizona mobility manager network is a joint effort of ADOT and the state's councils of governments (COGs) and metropolitan planning organizations (MPOs). While employed by regional public agencies, the positions are funded by ADOT grants using federal funds. They serve as advocates and coordinators of community transportation within their respective jurisdictions.

All the transportation providers interviewed expressed appreciation for the work of and support received by the mobility managers. While some had interest in even more support and active involvement, none expressed concern that the role of the mobility managers was taking the agency or the region served by the mobility manager in the wrong direction.

Implications for Technology

This finding demonstrates agency willingness to engage in coordination activities around which technology can be built. If the described receptivity to working with the mobility managers had been absent, it would be premature to explore technology outside of that which is dedicated to internal operations.

The finding also shows that the mobility manager network, while not currently focused on technology, can form a key foundation for future efforts that do have such a focus. Mobility managers themselves can develop greater knowledge where technology can be most relevant, and they can serve as an ongoing bridge between providers and a future state or regional resource that is centered on implementing technology.

Services Are Rooted in Community Needs

Why It Matters

While it is well outside the scope of this project to assess the appropriateness of the services to the communities they serve, a recurring theme discovered in the on-site interviews was how each agency showed signs of operating from a place of engagement with and understanding of the communities they serve.

What We Learned

The Freedom Express of Wickenburg operates without state or federal funding, and instead relies on local donations from riders, insurance coverage from the City of Wickenburg, and funding from both the Foundation for Senior Living and the Salvation Army

Through community engagement, Verde Valley Caregivers has been able to grow its budget from \$120,000 when Mr. Ellsworth became Executive Director nine years ago to a current \$800,000, one third of which comes from local donations.

New Horizons Disability Empowerment Center was able to build a new location for its services based on a donation of the site from a community member. Based on the needs of the people it serves who have high sensitivities to airborne chemicals, the new facility was designed with special building materials and equipped with air filtration units throughout.

Implications for Technology

Agencies that have a good understanding of the needs of their communities are in a much better position to establish sensible technology priorities. With so many possible directions to invest in, including rider-facing apps, in-vehicle technology, back-office tracking systems, and inter-agency coordination tools, those providers rooted in the reality of their communities are much less likely to be distracted by “shiny objects”, especially once they have been empowered with accurate information about the costs and benefits of those different tools.

Some Agencies Have Found Effective Technology Solutions

Why It Matters

Because of the specialized expertise needed, it can be exceedingly difficult for a small transportation provider to procure, implement, and maintain a technology system. Implementing a system that saves time, allows staff to better focus on advancing their mission, and supports growth should be noted, celebrated, and replicated wherever possible.

What We Learned

VICaP is using Ride Scheduler, a simple software as a service (SaaS) solution that fits that provider's type of operations with near perfection.

New Horizons Disability Empowerment Center (NHDEC) is using Easy Rides from GMV Syncromatics. The system is working well, despite some difficulties with in-vehicle tablets that have resulted in the need for some paper-based workarounds for drivers.

Verde Valley Caregivers Coalition (VVCC) is using Assisted Rides, a more complex SaaS solution that has required a significant investment of time and resources to implement. Though full implementation of Assisted Rides capabilities is still ongoing, the project is on track and roughly matches the Executive Director's original expectations for the level of effort required.

Implications for Technology

Together, the three examples above present three different common conditions that allow for success with technology deployments.

- **Simplicity**

As a small volunteer driver program, the operational needs of VICaP are relatively simple, and Ride Scheduler meets those needs with a simple solution. Ride Scheduler achieves this by limiting its functionality. By not trying to be the solution for every type of volunteer driver program and instead being designed around a few core operational assumptions (most notably assuming one rider per driver trip), assumptions that align with VICaP's concept of operations. The result is something that "just works".

- **Standardized Needs**

In terms of its operational profile, NHDEC is first and foremost a provider of demand-responsive non-emergency medical transportation (NEMT). While it certainly has its unique traits and culture, its concept of operations has enough similarity to the hundreds of other NEMT providers across the country that it was able to obtain technology through a relatively mature and competitive marketplace of solutions. In effect, NHDEC was able to hop on a train that was going in the right direction.

- **Robust Organizational Capacity**

VVCC is the most noteworthy of these three cases, because it has been able to implement technology in a domain (volunteer transportation) where the tools available in the marketplace are relatively immature, and implementation of the technology would significantly change the daily operations and culture of the organization to one that is more reliant on technology.

Fortunately, VVCC enjoys the leadership of Kent Ellsworth, a seasoned executive with an extensive background in leading organizations providing services (including

transportation) to vulnerable populations. Kent brought a bold vision for growth to VVCC that included a highly successful fundraising strategy which has allowed it to increase its budget by over 600% in the nine years of Mr. Ellsworth's tenure. This level of resources is a key factor supporting the organization's ability to take on a more sophisticated technology solution compared to VICaP.

When it came to evaluating the many options available, Mr. Ellsworth's experience, together with support from Mobility Manager Tod Morris at Northern Arizona Council of Governments, allowed him to develop realistic technical requirements and cost-benefit analyses to guide his selection of software. Remarkably, he has also been able to accurately estimate the amount of time and resources needed to roll out the solution, which included hiring temporary staff to assist in staff and driver training. As of this writing, nine months into deployment, VVCC is continuing to incrementally integrate the capabilities of Assisted Rides into daily operations.

Though the described solutions will not necessarily apply to other agencies with different operations, these successes point to a vendor marketplace that may be able to serve those agencies that do not yet have dedicated technology systems if the right conditions are in place.

Without diminishing the value of these victories, they should also be viewed as temporary, since maintenance issues, staff turnover, and poor end-of-life planning can undermine an otherwise successful implementation. Future research could include how the agencies plan to continue and build on their successes through budget and resource planning.

There is Strong Interest in Coordination and Development of a Statewide Vision and Network

Why It Matters

Coordination is hard work. Developing and maintaining coordination at a state or regional level, be it for service discovery, service design, or day-to-day operations, is not feasible without buy-in from transportation providers who must support it enough to dedicate time in their busy schedules and keep it alive.

What We Learned

In a manner that matched the positive perceptions of the mobility managers, all interviewees articulated a desire for more coordination in some form. Some examples:

- Multiple interviewees expressed interest in a single state-wide phone number to get access to mobility options.
- One mobility manager was interested in seizing opportunities to centralize scheduling and dispatching among multiple small providers to increase operational efficiency.
- One person at a transit agency wanted to see ADOT set up listservs to connect mobility professionals around the state.⁶³
- Multiple interviewees wanted to see expanded ability for providers to share knowledge and best practices.
- One small provider saw a need for a network that could provide more hands-on support for small providers, including a “toolkit for beginners” with such things as pre-made spreadsheets that allow a new provider to get off the ground quickly.
- One small urban transit provider wanted to see the development of a service network plan that values connectivity between rural communities.

Implications for Technology

The wide spectrum of possible focus areas for coordination means that there is no single technology solution. Possible directions can be broken into three broad groups, presented here in order of increasing effort required:

- Simple, highly targeted applications of technology focused on the dissemination of information among a small group. These tools can bring the highest gain with the lowest overhead. For connecting peers, this can be achieved with simple tools such as listservs, or more robust communication platforms such as Basecamp, which is used to connect mobility managers in the state of New York. For document sharing, there is a wide range of low-cost services that can allow a wealth of resources to be available at-hand throughout the state.
- Coordinated dissemination of rider-facing mobility information. These systems (frequently called “one-call/one-click” systems) can be carried out by a central, independent entity capable of carrying out the necessary outreach required for ongoing maintenance of the database. Age Friendly Arizona is currently developing just such a service in the form of a searchable listing of providers in the state. More robust versions, discussed in Track 2 of the recommendations of this report, include a trip planner.

⁶³ A listserv is an application that distributes messages to subscribers on an electronic mailing list. Yahoo Groups and Google Groups are current examples of this technology.

- The most sophisticated application of technology for coordination involves its use for integration of systems for day-to-day operations, such as centralized scheduling and dispatch. While it can result in service and efficiency improvements under a range of situations, such gains are very site-specific. Despite a large number of pilots around the country, there is no technology currently capable of reliably improving operations through coordination, independent of local conditions. In fact, the high cost of bringing coordination into operations at scale makes such efforts worthy of recommendation only when a solid foundation of institutional coordination is already in place and mature, and a concept of operations has been developed and thoroughly vetted by all stakeholders.

Given the current early stage of infrastructure for statewide coordination, the simplest technology is generally the best to start off with.

Maricopa Association of Governments (MAG) Has Capacity for Some State-Level Support

Why It Matters

Without some degree of state-level capacity to execute on technology initiatives, many of the most valuable uses of technology, namely those dedicated to service discovery, are simply not feasible. In addition, the application of technology strictly within an agency can be more successful with support from an umbrella entity that can bring resources for procurement, training, etc.

What We Learned

Age Friendly Arizona (AFA) demonstrates MAG's active engagement at a state level and is the means through which the agency is carrying out the Rural Transportation Initiatives project with state-wide reach. This kind of effort is not new to MAG, and there is an institutional commitment to continue it for the long haul. As part of its work under the AFA effort, MAG is deploying AZ Ride Info, a statewide directory of transportation services for older adults.

It is important that rural communities take the lead in this work as they know their communities the best, while MAG can leverage its capacity to serve in a supportive role. For this reason, Age Friendly Arizona is in the form of a partnership, with the roles of other partners brought to the fore.

Implications for Technology

MAG is willing and able to provide a range of support services to small transportation providers around the state.

System Weaknesses

Significant Isolation for Some Providers. The Smaller the Provider, the Greater the Isolation.

Why It Matters

While small providers are capable of doing much on their own, their capacity to serve their communities is likely to be hindered by a lack of connection with other similar organizations. This is particularly true with respect to technology, where the level of effort to put software solutions into place can be daunting.

What We Learned

The staff at one small volunteer driver program carried the misconception that they were unique as an organization, and that larger providers were the rule, even in rural areas. Their social isolation, while certainly not stopping them from creating and expanding their services over time, led to a more difficult learning curve in terms understanding what data to track about their services to attract more funding over time.

The staff at another volunteer driver program reported having minimal contact with its city's public transit service, despite the fact that a robust volunteer driver program would greatly benefit transit and the volunteer driver program could benefit from targeted support from the transit agency.

One small urban transit agency, while having the support of the city government to provide options within the city limits, feels isolated from the rest of the county in terms of access by transit. There is no connectivity with neighboring cities, and it's not possible to get to the county seat in one day by transit or intercity bus.

Implications for Technology

This finding serves as a counterpoint to the earlier finding of strong interest in coordination and development of a statewide vision and network. It is precisely because of systematic isolation that there is interest in more coordination. See the implications for technology for that finding for more details.

To the degree that the isolation relates to lack of service in place, then it is increased service, not technology, that is the most important factor to address. However, technology may be able to assist with a wealth of data when it comes to presenting information about service. A “one-click” system can increase visibility into all services by riders, planners, policymakers, and elected officials. Such systems make it easier to identify gaps in services and areas where new investments would bring the most benefit.

Regional Mobility Managers Do Not Have the Capacity to Develop State-Level Coordination

Why It Matters

If state-level coordination is a priority, then an effective organizational structure for mobility management must exist to achieve that outcome.

What We Learned

While the aforementioned positive regard for mobility managers is a clear strength in the current state system, that strength is largely limited to each manager's area of influence. Under guidelines of the state intelligent transportation systems (ITS) plan, each mobility manager has broad discretion to set their own priorities and apply technology in their own way. This finding is in no way a critique of the mobility managers or their work, but rather a recognition of limitations inherent in a structure of autonomous peers.

Implications for Technology

While much can be done with a committee-based approach to advancing coordination in the state, most technology initiatives require a more centralized structure with the ability to execute complex projects and maintain them over time.

There Needs to be a State-Level Entity with the Capacity to Support and Champion Implementation of Technology Best Practices

Why It Matters

State-level technology needs a sufficiently capable state-level actor. Efforts such as a statewide one-click system may be less sustainable if carried out by local or regional entities. A technical resource center to assist agencies with technology for operations would likely be more effective if it can gather information about the successes and challenges across the entire state rather than a single region.

What We Learned

ADOT, the most obvious candidate to spearhead state-wide specialized transit initiatives, has limited funding and resources to take on new significant projects. Further, the State of Arizona does not fund transit to a significant degree, and instead functions primarily as a pass-through for federal transit grants. Any attempt to initiate a technology project at ADOT requires very high-level approval that includes a “pitch” made to the Governor's office.

Implications for Technology

If a statewide vision for technology is to go beyond supporting regional initiatives, a suitable entity with appropriate and sustainable funding sources to carry out that vision must be found or established. While many elements of projects can and should be contracted out to vendors, there must be an entity that can develop a plan, oversee projects carried out through that plan, and be prepared to evolve the new infrastructure in a rapidly changing technology landscape.

External Opportunities

The Rural Transportation Incubator Project Is an Entry Point into Further Grant Opportunities

Why It Matters

Broadly speaking, a best practice for implementing technology is to start small and build on lessons learned. Age Friendly Arizona and its partners can make the case, starting with the RTI, for developing a new approach to solving rural mobility for Arizona.

What We Learned

There are two noteworthy impacts resulting from the RTI. One is the development of a network of stakeholders that ranges from very small providers to the state's largest urban MPO. The other is a linkage between that embryonic network and a national group of private philanthropists who have the potential to help further advance the goals of the incubator.

Implications for Technology

As reflected by the scarcity of research into the rural transit sector, technology for rural transit has been subject to consistent neglect and disinvestment since the point at which such technologies first became available. Some states have made inroads⁶⁴, but others have met with disappointing results stemming from the attempt to apply one-size-fits-all solutions to highly heterogeneous local conditions.⁶⁵ While ADOT's constraints limit its ability to participate in more robust solutions, it also allows for other possibilities generated from more philanthropic-based approaches. Possible benefits include a more iterative approach to discovering best practices and more resources allocated to developing networks of mutual support. Private funders may also have an increased interest in a more market-based approach to coordination such as the one outlined in AARP's report on universal mobility as a service.⁶⁶

⁶⁴ Two examples are Pennsylvania and Vermont. Pennsylvania has deployed a state-wide one-click system integrated with a scheduling and dispatch system used by most of its demand-response shared-ride program for senior citizens. More details can be found in the findings of the main report.

⁶⁵ One example is South Carolina. Curtis Sims, Program Manager at South Carolina DOT Office of Public Transit, recently presented on SCDOT's efforts to replace a statewide scheduling and dispatch procurement with best-fit solutions based on the needs of individual agencies. "Charting IT Procurement", presented at the National RTAP Technical Assistance Conference, September 17, 2019. Presentation slides available at

https://www.dropbox.com/sh/udhcxcar22osfyy/AAAMYI09il8Av9EJNnmqQbnga/Technology%20Presentations?dl=0&preview=Tech4+-+Charting+IT+Procurement+-+Curtis+Sims+and+Patricia+Drakeford.pdf&subfolder_nav_tracking=1

⁶⁶ Cited in the literature review of this report.

Software as a Service Eases Access to Technology Solutions

Why It Matters

There is ample evidence that successfully implementing technology is challenging and often beyond the reach of small organizations. Software as a Service (SaaS)⁶⁷ makes this work easier by reducing the barriers to hosting, maintaining, and paying for software.

What We Learned

As described earlier, both Verde Valley Caregivers and VICaP have successfully implemented SaaS solutions. Costs for both were at price points that were within reach of the agencies.

Implications for Technology

SaaS solutions increase small providers' access to technology by eliminating the burden of maintaining an on-premise computer server and, in many cases, by providing transparent pricing that an organization easily budget for. For example, costs may be based on the agencies number of system users, vehicles, trips, or other factors that can be easily forecast. With the proper gathering of requirements and thoughtful outreach to vendors, the success of Verde Valley Caregivers and VICaP is likely to be replicable to other similar small providers.

At the very smallest scale, where specialized software may not yet be a good fit, services such as G Suite and its online spreadsheet "Google Sheets" can offer opportunities for tools that make operations and coordination significantly easier. For instance, new agencies may share a common set of spreadsheet layouts and data manipulation scripts designed to meet all reporting requirements. While such an approach may scale poorly as an agency grows, it can get an operation off the ground quickly and cultivate learning about what data needs to be tracked. This understanding would make for a more knowledgeable user of specialized software when the time comes to graduate into a more robust system.

In terms of "one-click" systems for service discovery, all such systems in place are also SaaS, requiring no on-premise hosting or dedicated agency knowledge to maintain the software.

⁶⁷ Software as a service is a software licensing and delivery model in which software is licensed on a subscription basis, is centrally hosted, and generally accessed through a web browser. It contrasts with software with a license that is purchased once and installed onto on-premise computers by the licensee.

Technology Tools for Service Discovery Are More Affordable and Compelling Than in the Past

Why It Matters

Going from a relatively simple online directory of agencies to a tool that can provide trip planning is a significant step and may involve a great deal of foundational work in the areas of marketing and coordination. A lower cost on the software side can bring such an effort within reach.

What We Learned

Software costs for deploying an off-the-shelf open source one-click system starts at about \$25,000. This cost is enormously lower than the original development cost circa 2011, which was multiple millions of dollars. A more detailed presentation of estimated costs is in the cost analysis for recommendation track 2.

In addition, a new data standard, GTFS-flex, has been developed in recent years. An open source trip planner capable of consuming GTFS-flex data and giving trip plans that include demand-responsive services was developed for the state of Vermont and is now ready to be integrated into one-click software. The timeline for completion depends on pending grant applications but is estimated to be in the next 12-18.

Implications for Technology

There are a number of low-risk strategies that a statewide entity in Arizona could adopt to facilitate the discovery of mobility options. With this shift in the technology landscape, the risk for such projects shifts primarily to barriers related to institutional capacity and the maintainability of coordination schemes.

External Threats

Demographic trends and land use patterns

Why It Matters

We live in a society where technology is regularly presented as the central solution to solve daunting problems. In the case of rural transportation, that may be through the use of transportation network companies (TNCs), autonomous vehicles (AVs), or other proposed solutions.

What We Learned

Like many other parts of the United States, rural and exurban Arizona has distributed growth land use patterns combined with an aging population. In addition, several interviewees noted the particularly high proportion of retirement-age residents who moved to the state to retire and do not have extended family nearby. This combination of long distances, increasing need for mobility by people who cannot or should not drive, and thinner family support networks results in an unprecedented level of demand for transportation services. Volunteer driver programs report that their volunteer recruitment efforts often result in increased demand that outstrips the gains from the new volunteer drivers.

Implications for Technology

While technology can improve the accessibility and efficiency of transit and specialized transportation, it's important to acknowledge that technology can only marginally improve challenges that are primarily based on demographics and land use. TNCs, while a useful tool in urban and some suburban areas, are less likely to be effective in rural spaces. AVs, while a promising technology, are not ready for broad use and are unlikely to achieve readiness at scale for another decade or more.

Many Technology Solutions May Not Work Well with Target Demographic Groups

Why It Matters

Overly optimistic expectations for bringing technology directly to riders and drivers can result in projects that fall short of the desired outcomes.

What We Learned

We heard concerns from multiple providers that a large segment of riders would have difficulty using a mobile device or computer to access services. The landline telephone continues to be the baseline tools for communication for many riders.

In addition, it is this author's experience that many small rural transit agencies have found that driver-facing technology (e.g., tablets or mounted smartphones) are significantly more difficult to deploy and maintain than is portrayed by vendors. Vehicles are a challenging environment in which to keep devices in good working order, and the level of effort required to train drivers to use the tools effectively can be higher than agencies expect. The previously mentioned difficulties that New Horizons has experienced with its tablets matches with this general finding.

Implications for Technology

The promise of deploying connected technology tools to riders and drivers is very alluring. A single system that brings visibility to demand, supply, and dispatching resources can indeed bring significant efficiencies to a transportation service. Fully addressing this threat entails making thorough cost-benefit analyses which take into account the barriers described, the agency's capacity to implement and maintain the technology, and the organization's long-term strategic priorities as the technology and demographics shift over time.